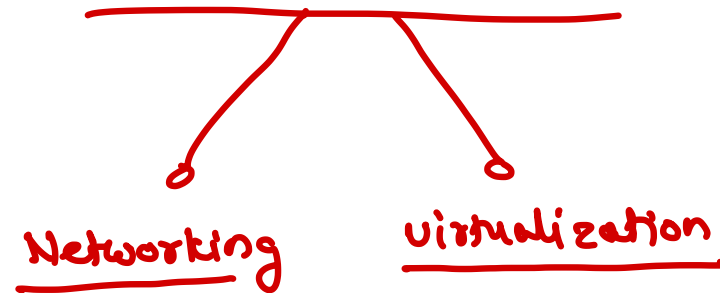




Cloud Computing

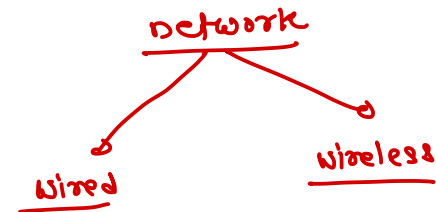
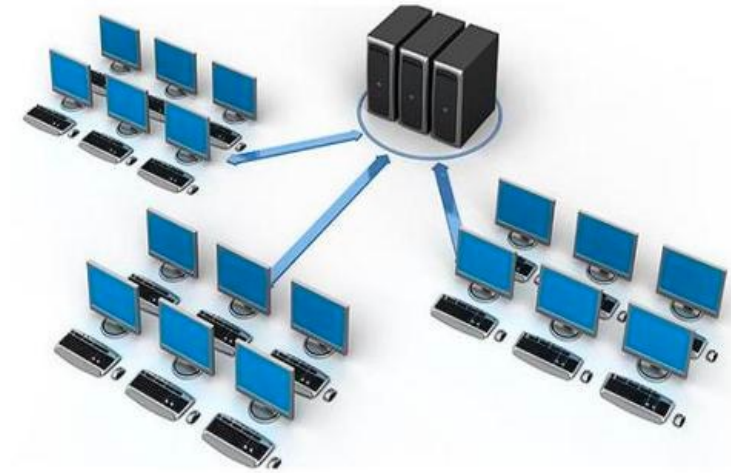
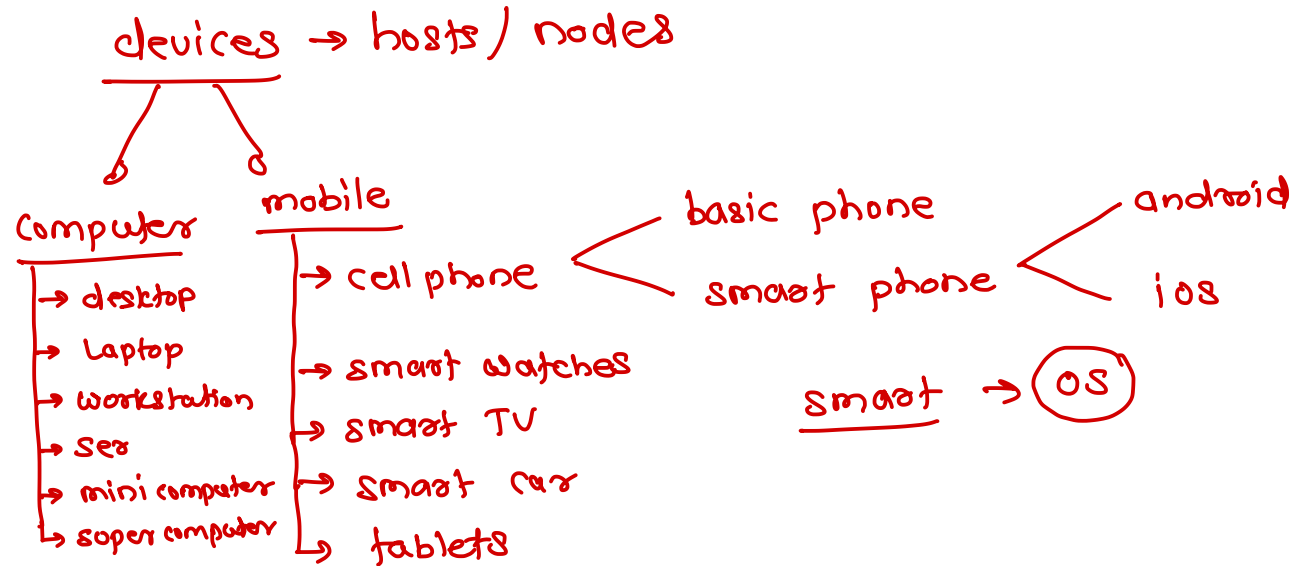


Networking



What is network ?

- It is the interconnection of multiple devices, generally termed as Hosts connected using multiple paths
- The purpose of network is: sending/receiving data or media → share data
- It involves various devices like hubs, switches, routers etc.



Wired network

- The network build by connecting devices together using wires/cables as a medium to transfer the data

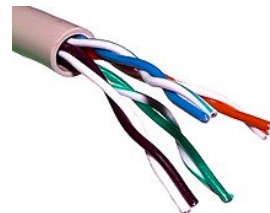
- Cables

- Coaxial cable
- Twisted pairs cables
- Fiber optics → light

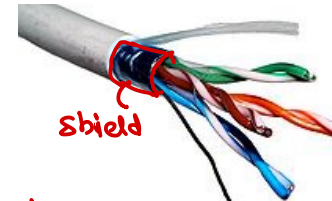
} electric signals

category
CAT cable
cat1
cat2
:
cat8

twisted pair cable



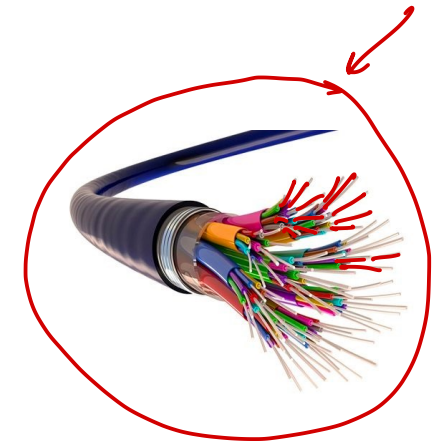
unshielded TP
(UTP)



shielded TP
(STP)

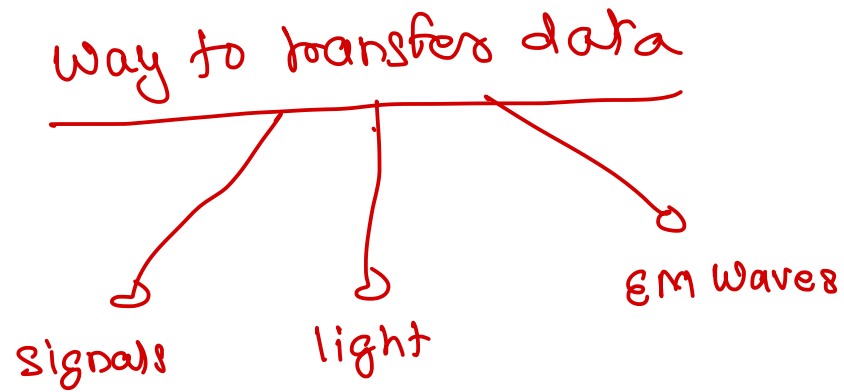


Fiber optics
single core
multi core



Wireless network

- The network build by connecting devices together using air as a medium to transfer the data
- EM Waves are used to transfer data from sender to receiver



Network Types

- Personal Area Network → wireless (PAN)
 - Smallest network which is very personal to the user
 - E.g. BlueTooth, NFC, IrDA
- Local Area Network wired
wireless - WLAN
 - Spans across building(s) and operated under single administrative system
 - E.g. company, school network
 - Technologies: Token Ring or Ethernet ✓
- Metropolitan Area Network (MAN)
 - Spans across cities
 - E.g. cable network
 - Technologies: high speed fiber optics
- Wide Area Network (WAN)
 - Spans across countries
 - Technologies: ATM, Frame Relay

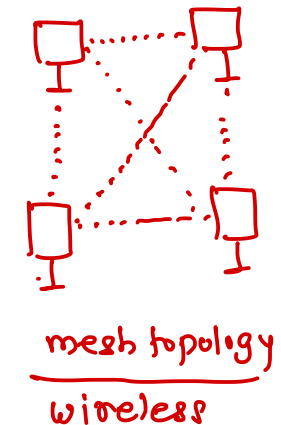
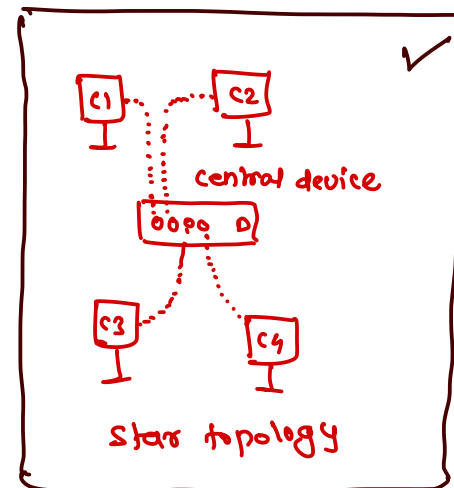
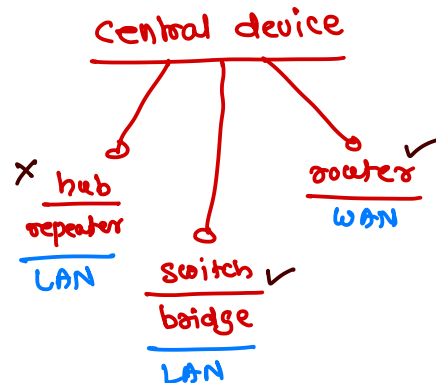
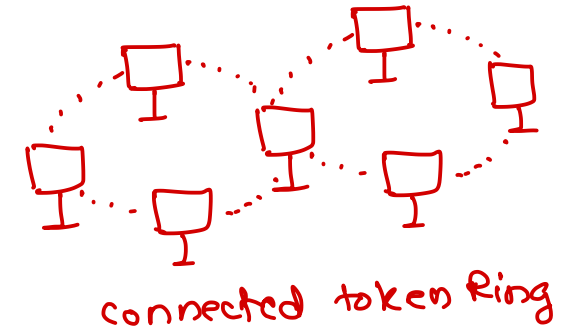
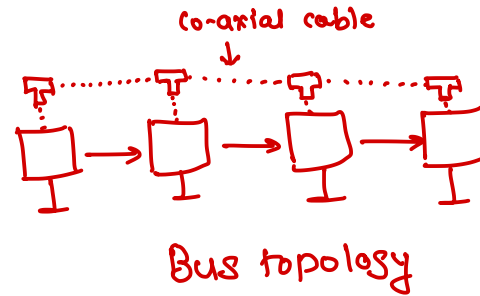


What is a network topology ?

- Physical arrangement of computers is known as topology

- Famous topologies

- Bus
- Ring
- Token Ring
- Star ✓
- Mesh



ISO OSI model

[Open System Interconnect] → layered model → reference model

- Conceptual model that characterizes and standardizes the communication functions of a telecommunication or computing system without regard to its underlying internal structure and technology
- Goal is the interoperability of diverse communication systems with standard communication protocols
- Layered architecture having 7 layers
 - Application
 - Presentation
 - Session
 - Transport
 - Network
 - Data Link
 - Physical

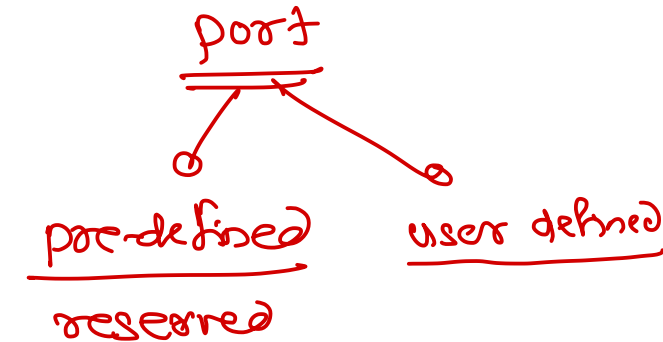
Standard - OSI
↓

implementation - TCP/IP



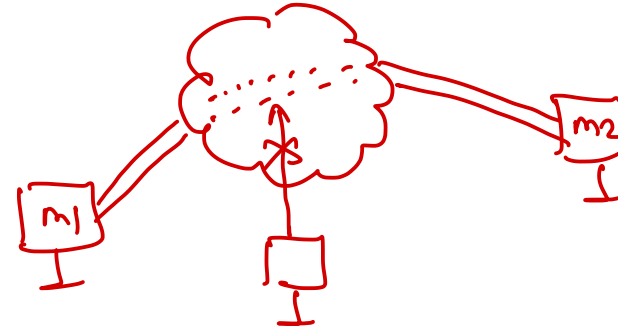
Application Layer 6 applications

- Specifies interface methods used by hosts in a communications network
- Contains communication protocols
 - HTTP [80]: Hyper Text Transfer Protocol
 - HTTPS [443]: Secure Hyper Text Transfer Protocol
 - FTP [20, 21]: File Transfer Protocol
 - SFTP [115]: Simple FTP
 - DNS [53]: Domain Name Service → translate domains to IP address
 - NFS [1023]: Network File System
 - POP3 [110]: Post Office Protocol
 - SMTP [25]: Simple Mail Transfer Protocol
 - SSH [22]: Secure Shell → communicate with remote machine using shell (commands)
 - LDAP [389]: Lightweight Directory Access Protocol



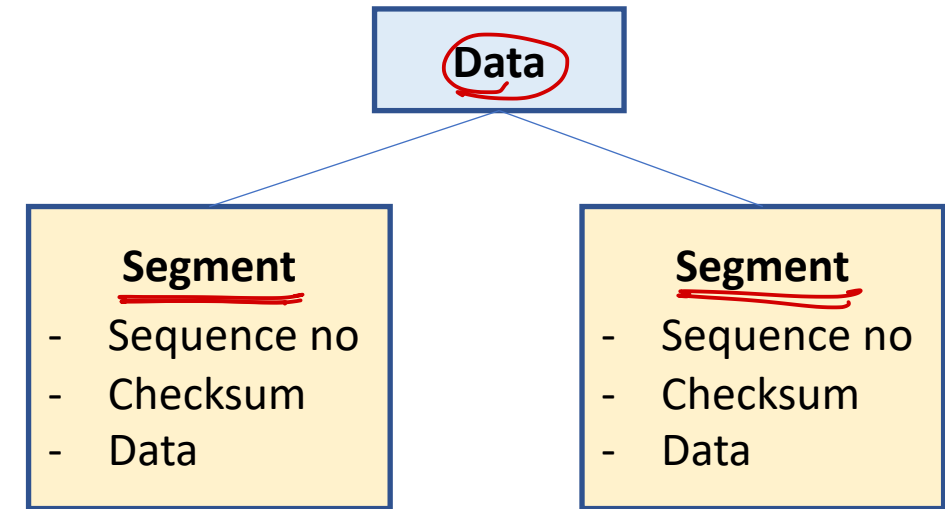
Session Layer

- Provides mechanism for opening, closing and managing session between processes
- Communication sessions consist of requests and responses that occur between applications
- Protocols
 - ASP: AppleTalk Session Protocol ✗
 - ADSP: AppleTalk Data Stream Protocol ✗
 - NetBIOS: Network BIOS
 - PAP: Password Authentication Protocol
 - PPTP: Point to Point Tunneling Protocol } VPN
 - RPC: Remote Procedure Call
 - SCP: Session Control Protocol } manage session
 - SDP: Socket Direct Protocol



Transport Layer

- Provide host-to-host communication services for applications
- Creates Segment (data unit) containing
 - Sequence number :
 - Checksum - CRC → error checking
 - Port number
- Protocols
 - TCP
 - Connection oriented protocol
 - Provides: Flow Control, Error checking, windowing
 - Guarantees data delivery
 - Slower than UDP
 - E.g. WWW, HTTP
 - UDP
 - Connectionless protocol
 - Does not provide flow control
 - Does not guarantee data delivery
 - Faster than TCP
 - E.g. streaming, online games

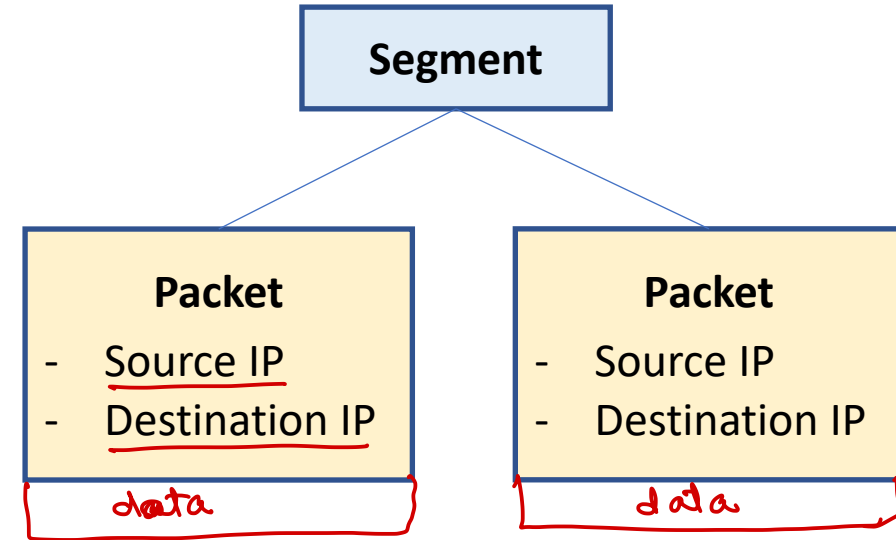
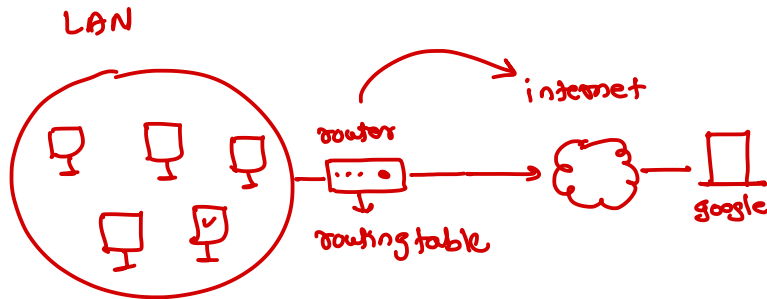


Network Layer

IP address

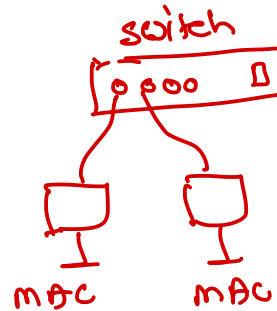
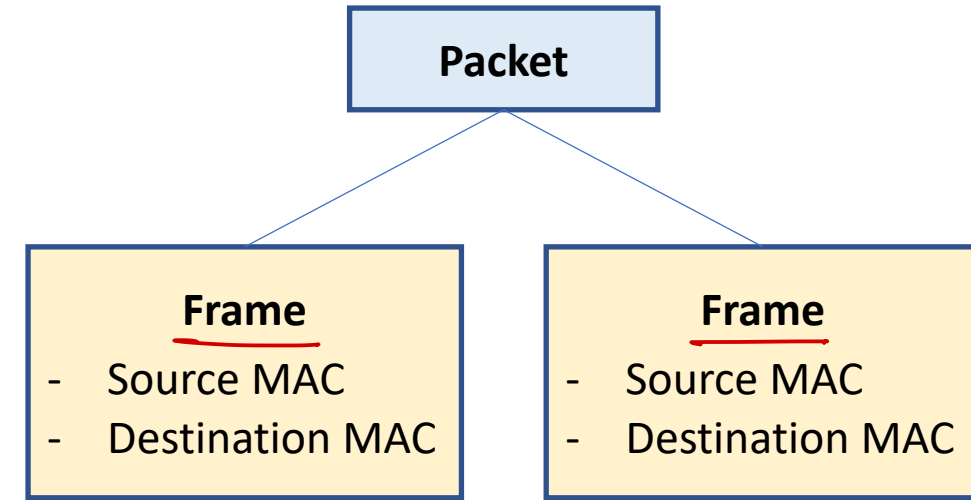
- Responsible for packet forwarding including routing through intermediate routers
- Responsible for splitting segment into packets containing
 - Source IP address
 - Destination IP address
- Protocols
 - IP: Internet Protocol
 - IPX: Internetwork Packet Exchange
 - IPSec: Internet Protocol Security
 - EGP: Exterior Gateway Protocol

BGP : Border → — — — —



Data Link Layer 8 mac address

- Transfers data between
 - adjacent network nodes in a wide area network (WAN) or
 - between nodes on the same local area network (LAN) segment
- Encapsulates packet into Frames containing
 - Source MAC Address
 - Destination MAC Address
- Sublayers
 - Logical Link Layer
 - Media Access Control Layer



MAC - Media Access Control



Data Link Layer: Logical Link Layer

- The uppermost sublayer multiplexes protocols running at the top of data link layer, and optionally provides flow control, acknowledgment, and error notification
- Provides addressing and control of the data link
- Services
 - Error control (automatic repeat request, ARQ)
 - Flow control [Data-link-layer flow control is not used in LAN protocols such as Ethernet, but in modems and wireless networks]



Data Link Layer: Media Access Control Layer

- Refers to the sublayer that determines who is allowed to access the media at any one time (CSMA/CD)
- Determines where one frame of data ends and the next one starts (frame synchronization)
- Frame synchronization uses: time based, character counting, byte stuffing and bit stuffing.
- Services
 - Multiple access protocols for channel-access control,
 - CSMA/CD protocols for collision detection and re-transmission in Ethernet networks
 - CSMA/CA protocol for collision avoidance in wireless networks
 - Physical addressing (MAC addressing)
 - LAN switching (packet switching), including MAC filtering, Spanning Tree Protocol (STP) and Shortest Path Bridging (SPB)
 - Data packet queuing or scheduling



Physical Layer

- Consists of the electronic circuit transmission technologies of a network
- Fundamental layer underlying the higher level functions in a network which provides means of transmitting raw bits rather than logical packets or segments
- The bitstream may be grouped into code words or symbols and converted to a physical signal that is transmitted over a transmission medium wire / air
- Translates logical communications requests from the data link layer into hardware-specific operations to cause transmission or reception of electronic signals
- Services
 - Modulation/Demodulation
 - Multiplexing
- Consists of
 - Cables/wires
 - Devices like hub, repeaters etc.



application PDU H data - application

presentation PDU H H data → presentation

session PDU H H H data → session

segment H H n H data → transport

packet H H H H H data → network

frame H H H n H n data → data link

bitstream H H H H H H H data → physical



010001001110001110000110

Addressing Modes: MAC Address

Given by manufacturer

- Used to identify NIC uniquely
- Consists of 6 bytes [48 bits]
- First 3 bytes represents manufacturer
- Next 3 bytes represents NIC's unique address

mac address → physical address → Read only addressed



Addressing Modes: IP Address → logical address

- Used to identify every device uniquely
- Set by operating system running on the device
- Can be written in
 - Decimal: 192.168.100.10 → human
 - Binary: 11000000.10101000.01100100.00001010 → computer
- Types of IPv4
 - Private: used to communicate with other devices in local network
 - Public: used to communicate with other devices over internet ←
- Versions
 - IPv4
 - 32 bit [4 bytes] address
 - Classful and Classless addressing
 - IPv6
 - 128 bit address



IP classes

A - 0 - 127
0.0.0.0 - 127.255.255.255 → 127.x.x.x → 10.x.x.x
loopback private

B - 128 - 191 → 172.16.0.0 to 172.31.255.255
private

C - 192 - 223 → 192.168.x.x
private

D → 224 - 239 reserved for multicasting

E → 240 - 255 reserved for RnD

Virtualization



Software Deployment

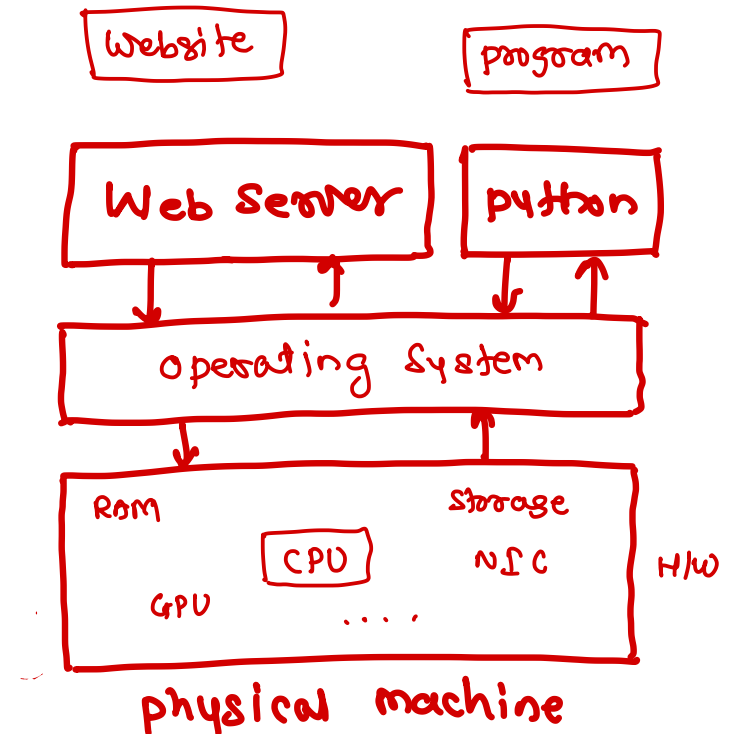
- Software deployment defines a process of making the software available for the users
- E.g. a web site can be available for the end users, when it is hosted on a machine which can be accessed from anywhere in the world

Copied
deployed



Traditional Deployment : physical machines , slow provision

- Early on, organizations ran applications on physical servers
- There was no way to define resource boundaries for applications in a physical server, and this caused resource allocation issues
- For example, if multiple applications run on a physical server, there can be instances where one application would take up most of the resources, and as a result, the other applications would underperform
- A solution for this would be to run each application on a different physical server
- But this did not scale as resources were underutilized, and it was expensive for organizations to maintain many physical servers



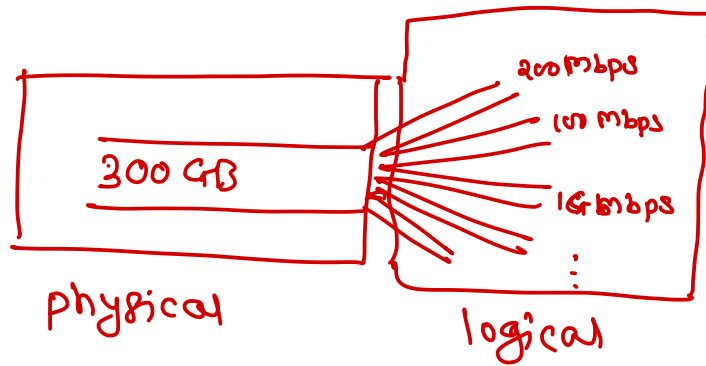
What is virtualization

- Virtualization is the creation of a virtual -- rather than actual -- version of something, such as an operating system (OS), a server, a storage device or network resources
- Virtualization uses software that simulates hardware functionality in order to create a virtual system
- This practice allows IT organizations to operate multiple operating systems, more than one virtual system and various applications on a single server
- Types
 - ✓ ■ Network virtualization
 - Storage virtualization
 - Data virtualization
 - Desktop virtualization
 - Application virtualization
 - Hardware virtualization ✕ ✕ ✕



Network Virtualization ✓

- Network virtualization takes the available resources on a network and breaks the bandwidth into discrete channels
- Admins can secure each channel separately, and they can assign and reassign channels to specific devices in real time
- The promise of network virtualization is to improve networks' speed, availability and security, and it's particularly useful for networks that must support unpredictable usage bursts



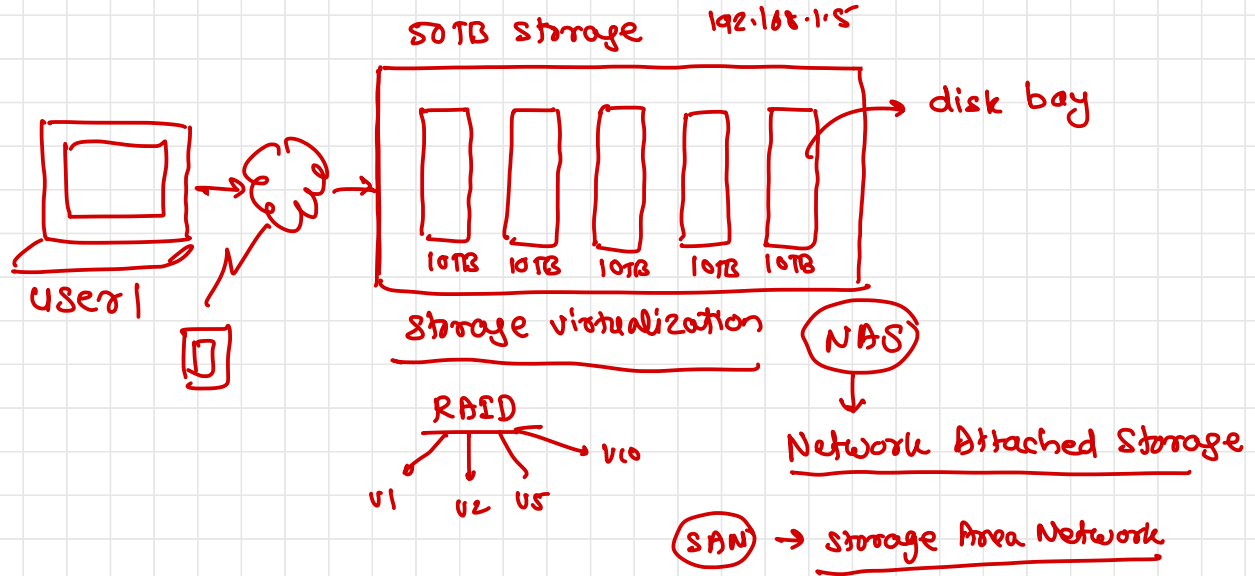
Storage Virtualization

- Storage virtualization is the pooling of physical storage from multiple network storage devices into what appears to be a single storage device that is managed from a central console
- Storage virtualization is commonly used in storage area networks
- Applications can use storage without having any concern for where it resides, what technical interface it provides, how it has been implemented, which platform it uses and how much of it is available
- Benefits
 - Makes the remote storage devices appear local
 - Multiple smaller volumes appear as a single large volume
 - Data is spread over multiple physical disks to improve reliability and performance
 - All operating systems use the same storage device
 - Provided high availability, disaster recovery, improved performance and sharing



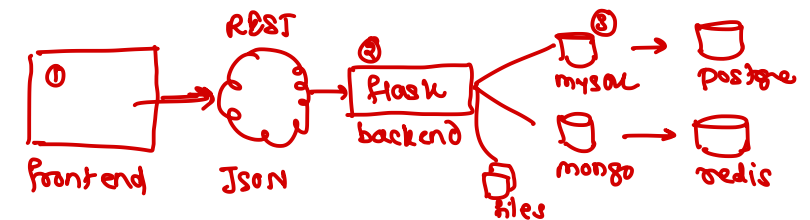
40TB → 10TB + 10TB + 10TB + 10TB + 10TB

12TB



Data virtualization

- Data virtualization is the process of aggregating data from different sources of information to develop a single, logical and virtual view of information so that it can be accessed by front-end solutions such as applications, dashboards and portals without having to know the data's exact storage location
- The process of data virtualization involves abstracting, transforming, federating and delivering data from disparate sources
- The main goal of data virtualization technology is to provide a single point of access to the data by aggregating it from a wide range of data sources
- Benefits
 - Abstraction of technical aspects of stored data like APIs, Language, Location, Storage structure
 - Provides an ability to connect multiple data sources from a single location
 - Provides an ability to combine the data result sets across multiple sources (also known as data federation)
 - Provides an ability to deliver the data as requested by users



Desktop virtualization

- With desktop virtualization, the goal is to isolate a desktop OS from the endpoint that employees use to access it
- It provides an ability to connect to the desktop from remote site
- When multiple users connect to a shared desktop, as is the case with Microsoft Remote Desktop Services, it's known as shared hosted desktop virtualization

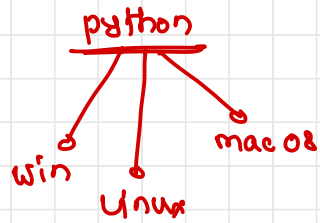
Sharing desktop across network



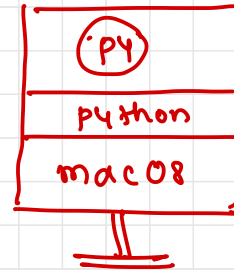
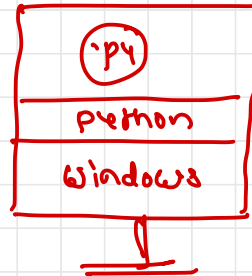
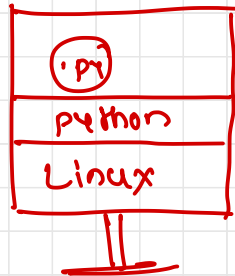
Application Virtualization

- With application virtualization, an app runs separately from the device that accesses it
- Application virtualization makes it possible for IT admins to install, patch and update only one version of an app rather than performing the same management tasks multiple times





①



physical machines

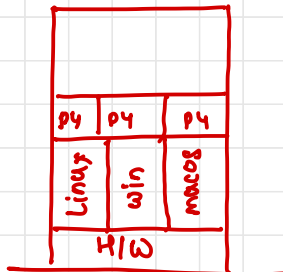
→ high cost

→ faster

②

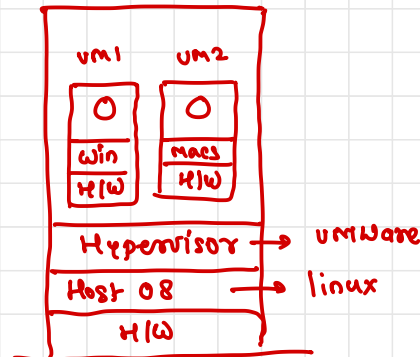
multi-os installation

multi booting



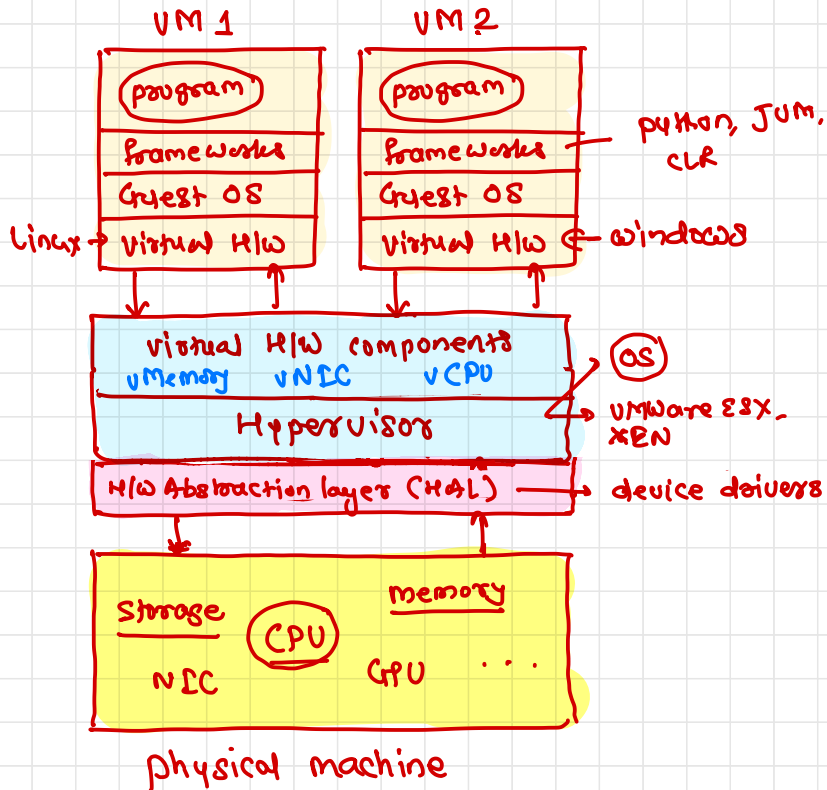
③

H/W virtualization



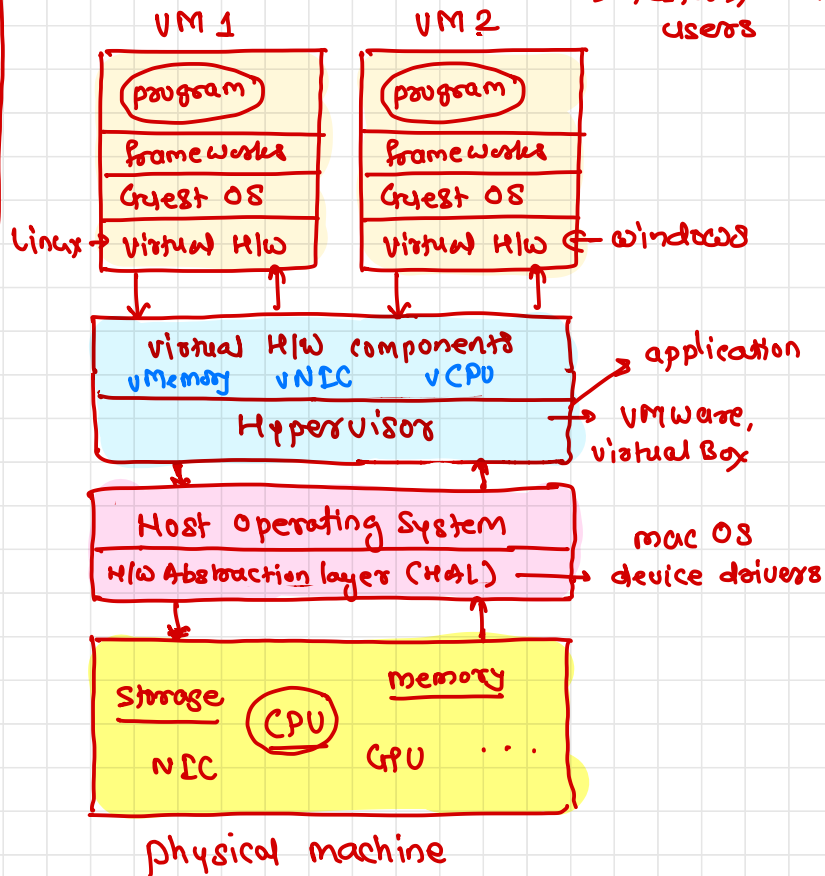
Type I H/W virtualization

- Faster, better H/W utilization
↳ cloud providers



Type II H/W virtualization

↳ testers, end users



Hardware Virtualization

- Hardware virtualization or platform virtualization refers to the creation of a virtual machine that acts like a real computer with an operating system
- The process of masking the hardware resources like
 - CPU
 - Storage
 - Memory
- For example, a computer that is running Microsoft Windows may host a virtual machine that looks like a computer with the Ubuntu Linux operating system; Ubuntu-based software can be run on the virtual machine
virtual
- The process of creating Machines
^



Virtual Machine

- A virtual machine is the emulated equivalent of a computer system that runs on top of another system
- Virtual machines may have access to any number of resources
 - Computing power - through hardware-assisted but limited access to the host machine's CPU
 - Memory - one or more physical or virtual disk devices for storage
 - A virtual or real network interfaces
 - Any devices such as
 - video cards,
 - USB devices,
 - other hardware that are shared with the virtual machine
- If the virtual machine is stored on a virtual disk, this is often referred to as a disk image



Types of hardware virtualization

■ Type I — no host OS

- A Type 1 hypervisor runs directly on the host machine's physical hardware, and it's referred to as a bare-metal hypervisor
- It doesn't have to load an underlying OS first
- With direct access to the underlying hardware and no other software, it is more efficient and provides better performance
- It is best suited for enterprise computing or data centers
- E.g. VMware ESXi, Microsoft Hyper-V server and open source KVM

■ Type II

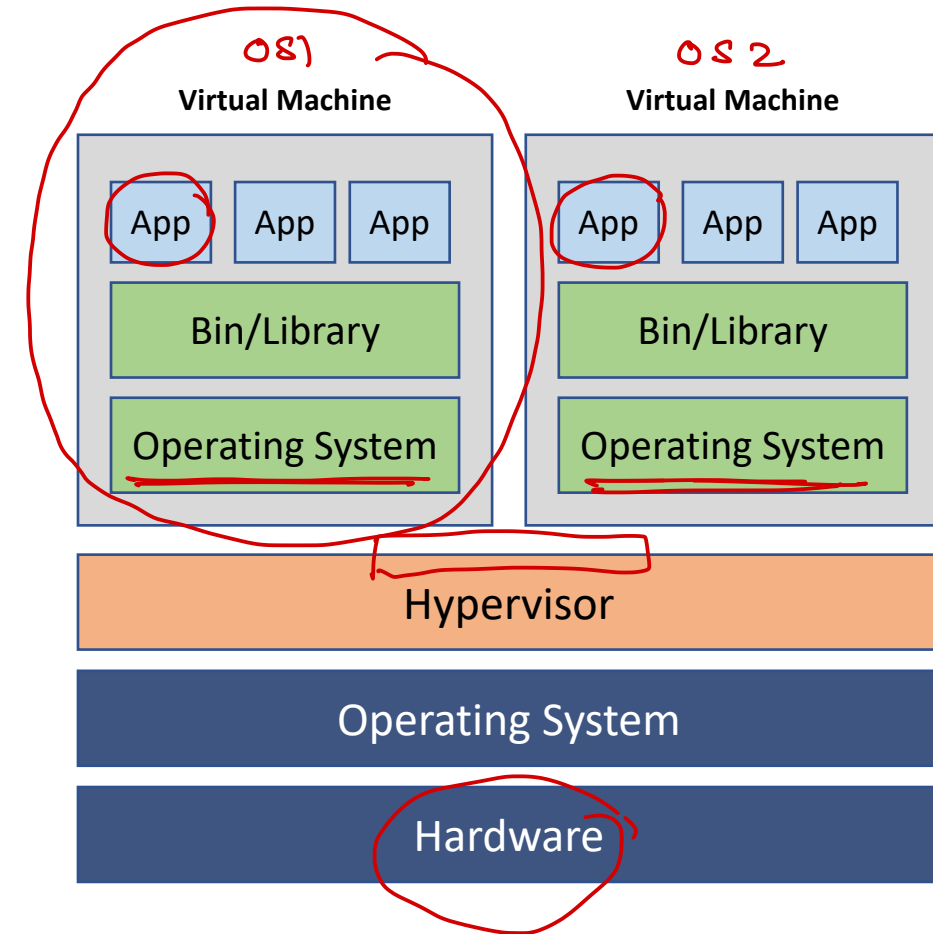
host OS

- A Type 2 hypervisor is typically installed on top of an existing OS, and it's called a hosted hypervisor
- It relies on the host machine's pre-existing OS to manage calls to CPU, memory, storage and network resources → HAL
- E.g. VMware Fusion, Oracle VM VirtualBox, Oracle VM Server for x86, Oracle Solaris Zones, Parallels and VMware Workstation



Virtualized Deployment

- It allows you to run multiple Virtual Machines (VMs) on a single physical server's CPU
- Virtualization allows applications to be isolated between VMs and provides a level of security as the information of one application cannot be freely accessed by another application
- Virtualization allows better utilization of resources in a physical server and allows better scalability because
 - an application can be added or updated easily
 - reduces hardware costs
- With virtualization you can present a set of physical resources as a cluster of disposable virtual machines
- Each VM is a full machine running all the components, including its own operating system, on top of the virtualized hardware



Advantages of virtualization

- **Lower costs**

- Virtualization reduces the amount of hardware servers necessary within a company and data center
- This lowers the overall cost of buying and maintaining large amounts of hardware

- **Easier disaster recovery**

- Disaster recovery is very simple in a virtualized environment
- Regular snapshots provide up-to-date data, allowing virtual machines to be feasibly backed up and recovered
- Even in an emergency, a virtual machine can be migrated to a new location within minutes

- **Easier testing**

- Testing is less complicated in a virtual environment
- Even if a large mistake is made, the test does not need to stop and go back to the beginning
- It can simply return to the previous snapshot and proceed with the test.



Advantages of virtualization

- **Quicker backups**

- Backups can be taken of both the virtual server and the virtual machine
- Automatic snapshots are taken throughout the day to guarantee that all data is up-to-date
- Furthermore, the virtual machines can be easily migrated between each other and efficiently redeployed

- **Improved productivity**

- Fewer physical resources results in less time spent managing and maintaining the servers
- Tasks that can take days or weeks in a physical environment can be done in minutes
- This allows staff members to spend the majority of their time on more productive tasks, such as raising revenue and fostering business initiatives



Cloud

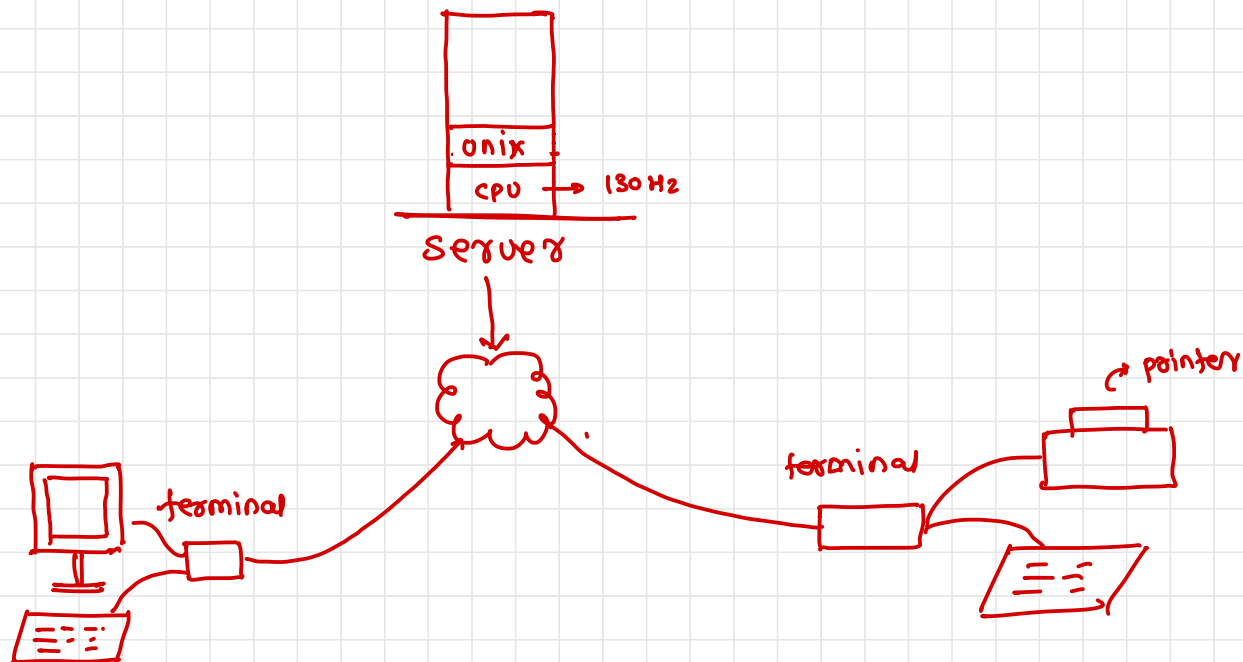


Computing Model

- Desktop computing
- Client-Server computing
- Cluster computing
- Grid computing ✖
- Cloud Computing ✓

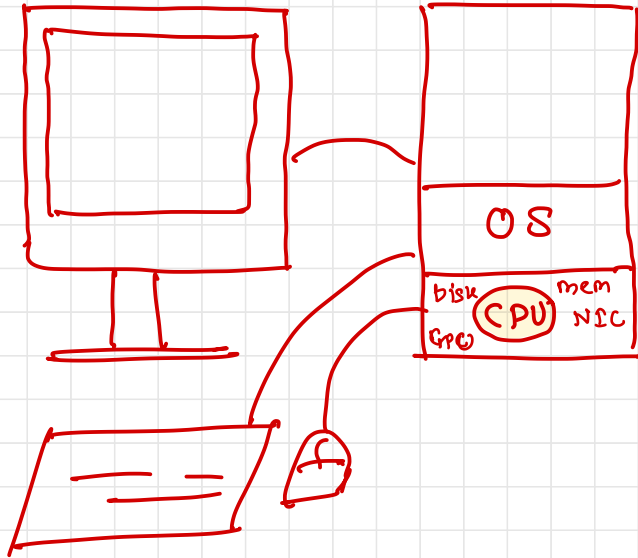


① pre-desktop

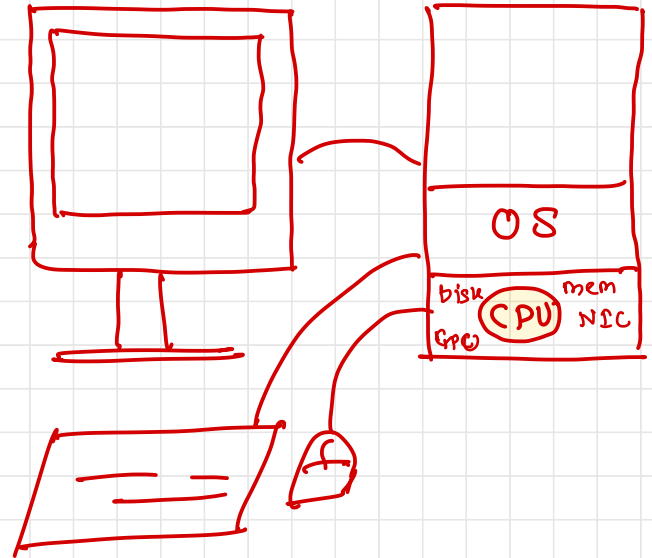


① Desktop computing

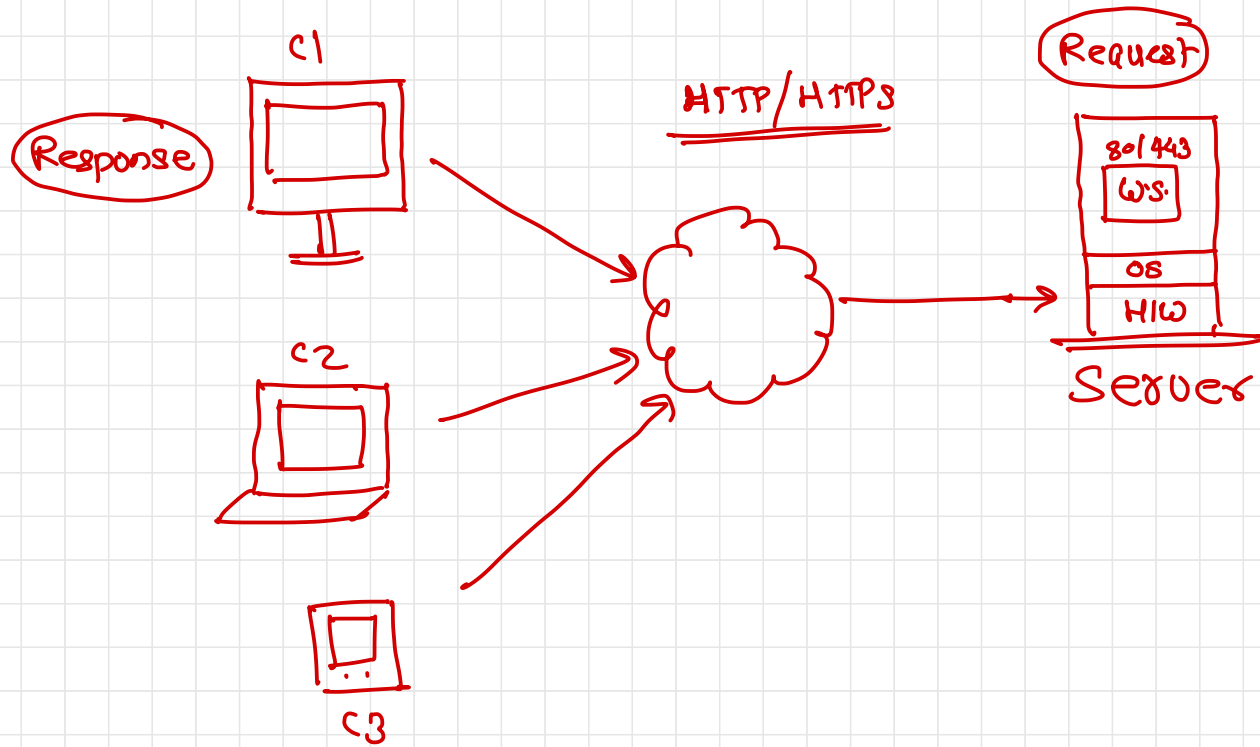
user 1



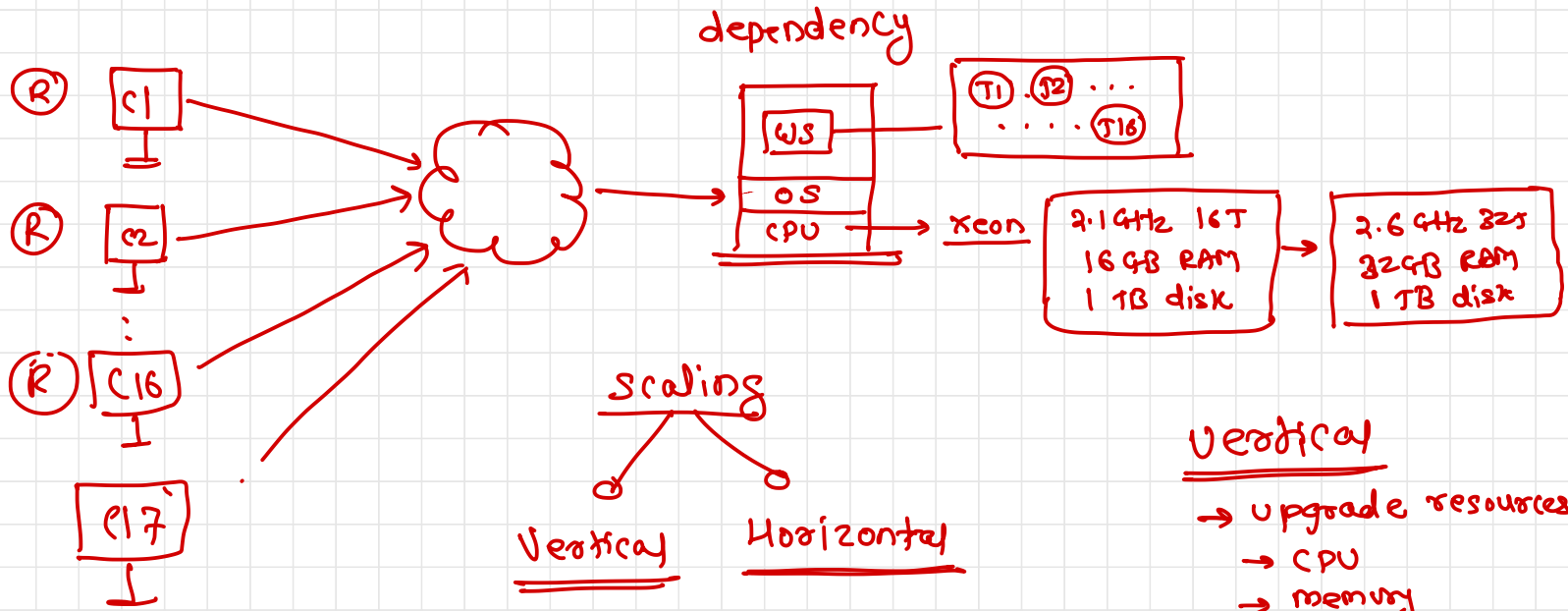
user 2



a) client-server computing



3.1 client-server



Vertical

- upgrade resources
 - CPU
 - memory
 - disk
- limitations of H/W
- dependency
- static / known load

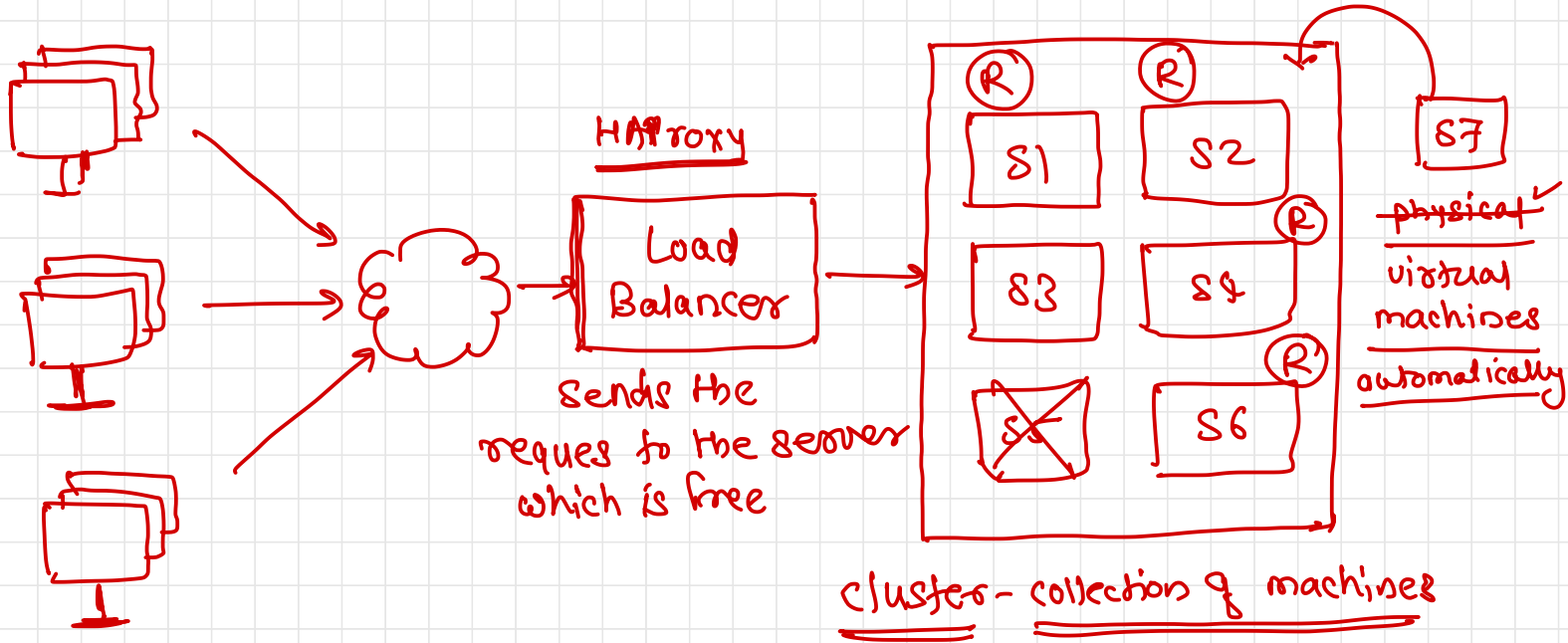
④ cluster computing

System Design
scalable system

Highly Available App
(HA)

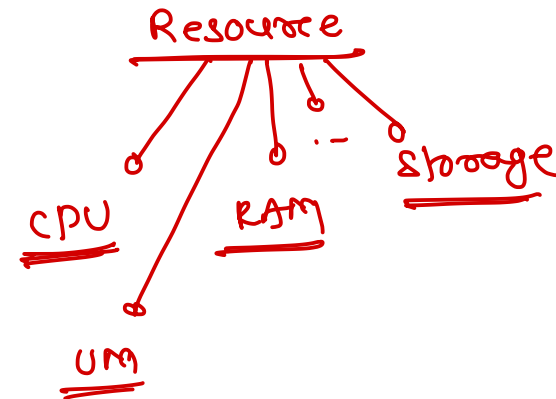
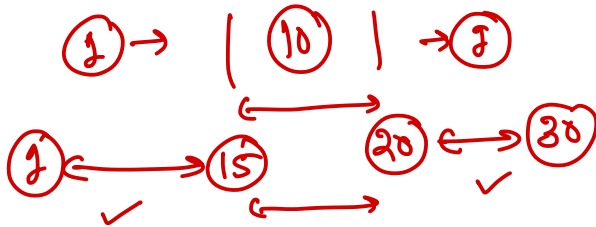
Horizontal Scaling

- clone the servers & create cluster of servers
- dynamic load



What is cloud computing ?

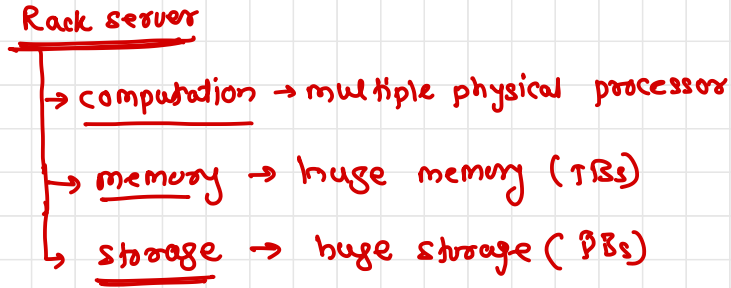
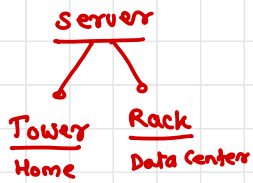
- The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer.
- Is the delivery of on-demand computing resources – everything from data centers over the internet on a pay for use basis
- Cloud computing is an umbrella term used to refer to Internet based development and services



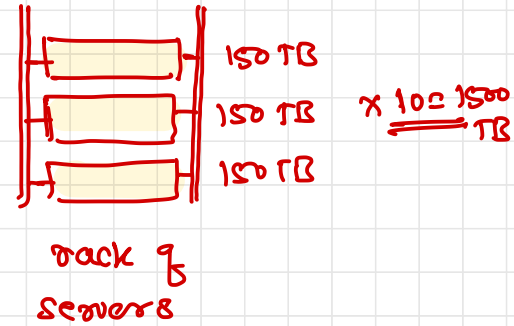
What is Data Center ?

- Where your IT devices and applications are located
- For a non-technical person it is the cloud where the user's files/data is stored
- Components
 - Servers
 - Security
 - WAN
 - Storage
 - File Sharing





>



→ rack server

→ racks of servers

→ row of racks

→ room filled with rows of racks

→ floor of multiple rooms

→ building of multiple floors

→ data center having multiple buildings

→ availability zone of at least two data centers

→ region of multiple AZs

→ provider having multiple regions

→ many cloud providers

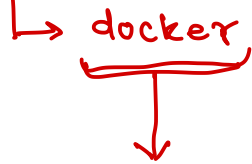
What is Virtualization ?

- Refers to the act of creating a virtual (rather than actual) version of something, including virtual computer hardware platforms, storage devices, and computer network resources

- Types

- Type I ✓
 - Type II ✓
- } hardware virtualization

- Containerization → OS virtualization



→ faster
low-resource
requirements > than vms

container orchestration

Kubernetes, Marathon,
Mesos, Docker Swarm



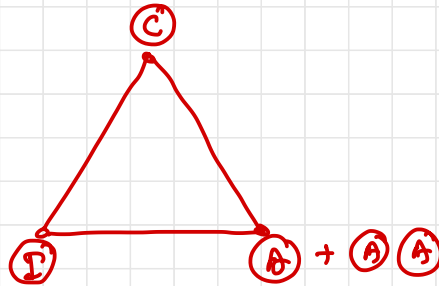
Terminologies

- **Scalability** → *scalable architecture → can expand or shrink*
 - refers to the idea of a system in which every application or piece of infrastructure can be expanded to handle increased load
- **Elasticity** ← *on-demand resources* →
 - the degree to which a system is able to adapt to workload changes by provisioning and de-provisioning resources in an *autonomic manner*, such that at each point in time the available resources match the current demand as closely as possible
- **Availability**
 - refers to the ability of a user to access information or resources in a specified location and in the correct format
- **Information Assurance**
 - *availability*, *integrity*, *authentication*, *confidentiality* and *nonrepudiation*
- **On-demand service**
 - A model by which a customer can purchase cloud services as needed



Security

- authentication → validating user [user name & password]
- authorization → allowing authenticated user to use certain parts of app
- confidentiality → keeping confidential information confidential → encryption
passwords
- integrity → getting valid information all the time
- availability → making app available to end users



Service Models

infrastructure - resources
CPU / memory / storage

public

Private Cloud

Applications

Data

OS

Virtualization

Servers

Storage

Networking

Customer

infrastructure

virtual → operations

Infrastructure as a Service

Applications

Data

OS

Virtualization

Servers

Storage

Networking

Customer

Provider

Developers

Platform as a Service

Applications

Data

OS

Virtualization

Servers

Storage

Networking

Customer

Provider

end users

Software as a Service

Applications

Data

OS

Virtualization

Servers

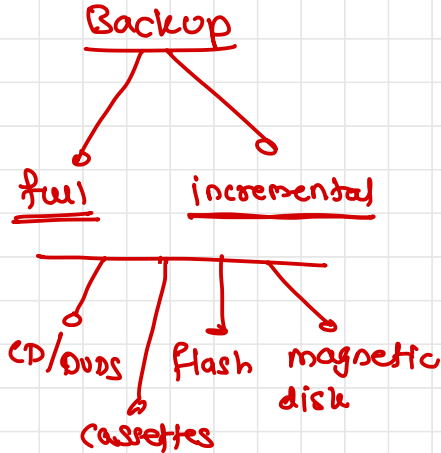
Storage

Networking

Provider

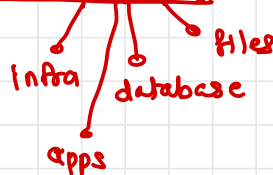
Development

→ SCM → Source Code Management
Revision Control System → git, cvs, svn



Service model

- IaaS - Infrastructure -
- PaaS - Platform -
- SaaS - software -
- **Daas - Database** -
- FaaS - Functions -
- EaaS - Everything -



99.999999999

SLA

Service Models

Software
as a Service (SaaS)

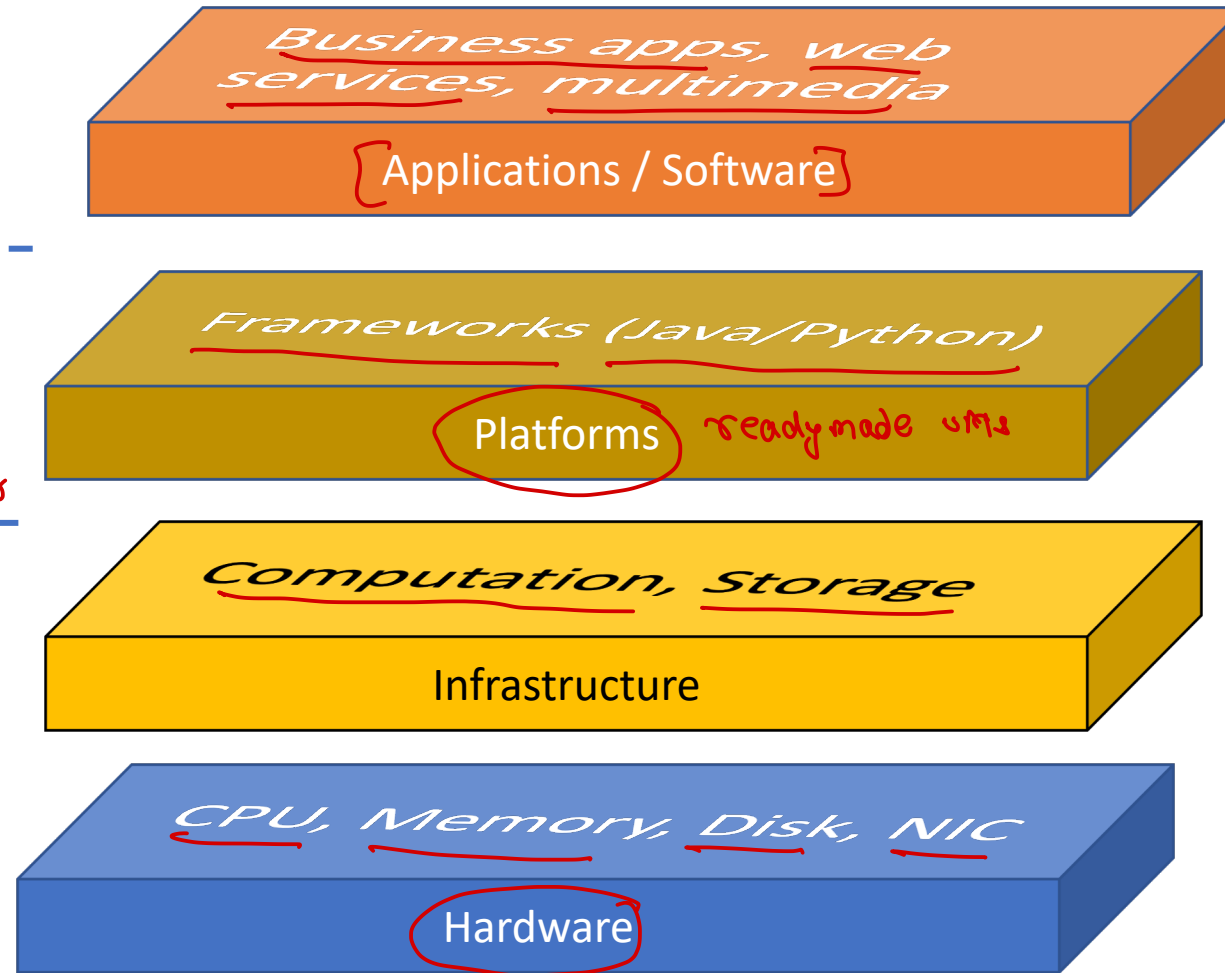
end user

Platform
as a Service (PaaS)

developers

Infrastructure
as a Service (IaaS)

operations team



Google Apps,
Facebook, YouTube,
Dropbox, Google Photos

Google App Engine,
Amazon Simple DB, S3,
Microsoft Azure

Amazon EC2,
Google Compute VM,
Azure VM

Data Center



Service Models: IaaS

- Infrastructure as a Service
- Allocates virtualized computing resources to the user through the internet
- IaaS is completely provisioned and managed over the internet
- helps the users to avoid the cost and complexity of purchasing and managing their own physical servers
- Every resource of IaaS is offered as an individual service component and the users only have to use the particular one they need → vm, storage, Database
- The cloud service provider manages the IaaS infrastructure while the users can concentrate on installing, configuring and managing their software
- Generally meant for operations team to setup the required infrastructure
- Benefits
 - Time and cost savings: more installation and maintenance of IT hardware in-house,
 - Better flexibility: On-demand hardware resources that can be tailored to your needs,
 - Remote access and resource management.

↪ management console



Service Models: PaaS

- Provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure typically associated with developing and launching an app
- Generally meant for developers
- Benefits
 - Mastering the installation and development of software applications
 - Time saving and flexibility for development projects: no need to manage the implementation of the platform, instant production
 - Data security: You control the distribution, protection, and backup of your business data



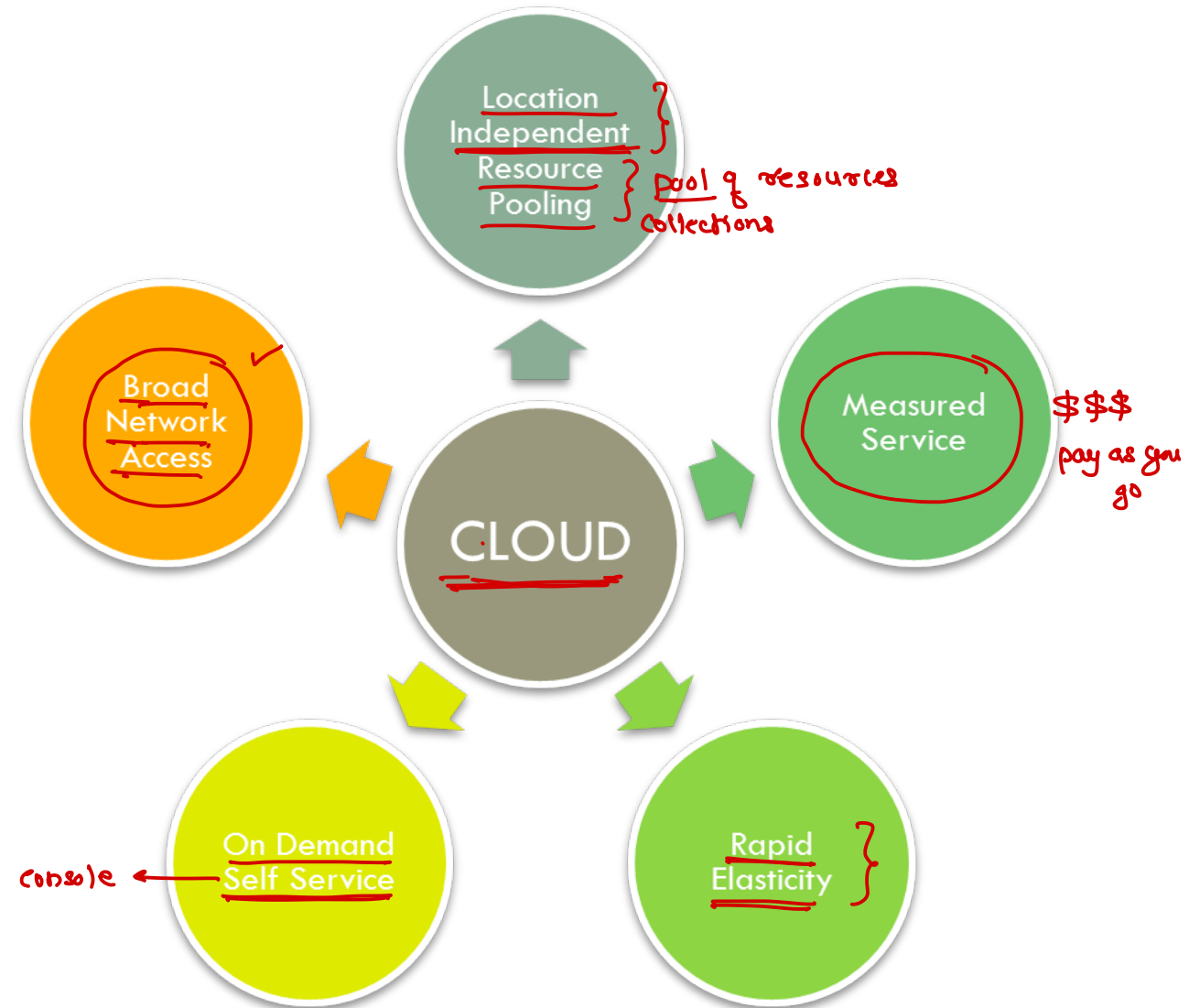
Service Models: SaaS

- Software as a Service
- Software distribution model in which a third-party provider hosts applications and makes them available to customers over the Internet
- User won't know which computer or operating system or infrastructure is used to host the software
- Generally meant for end user
- Benefits
 - You are entirely free from the infrastructure management and aligning software environment: no installation or software maintenance
 - You benefit from automatic updates with the guarantee that all users have the same software version
 - It enables easy and quicker testing of new software solutions.



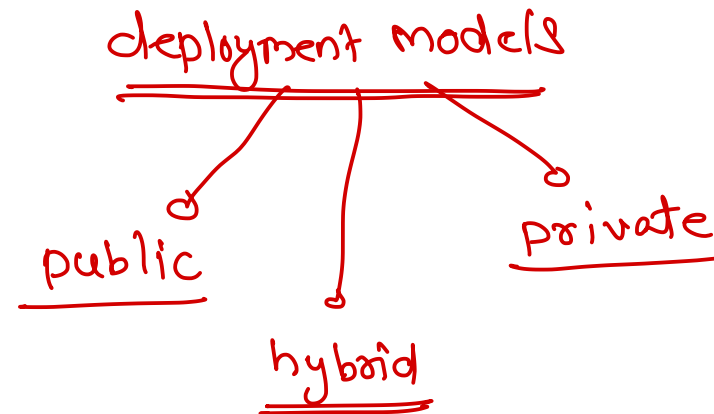
Cloud Computing Characteristics

- Rapid Elasticity
- On Demand Self Service
- Broad Network Access
- Location Independent Resource Sharing
- Measured Services



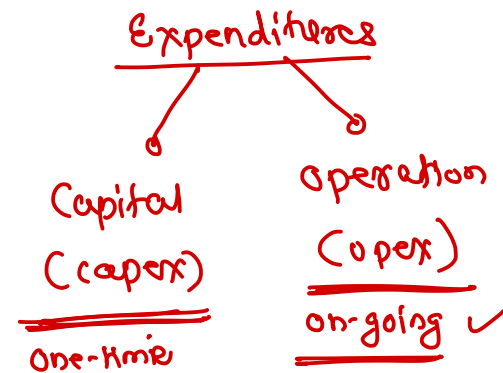
Cloud Deployment Models: Public - internet ✓

- Supports all users who want to make use of a computing resource, such as hardware (OS, CPU, memory, storage) or software (application server, database) on a subscription basis
- Most common uses of public clouds are for application development and testing, tasks such as file-sharing, and e-mail service
- Requires internet to access the resources



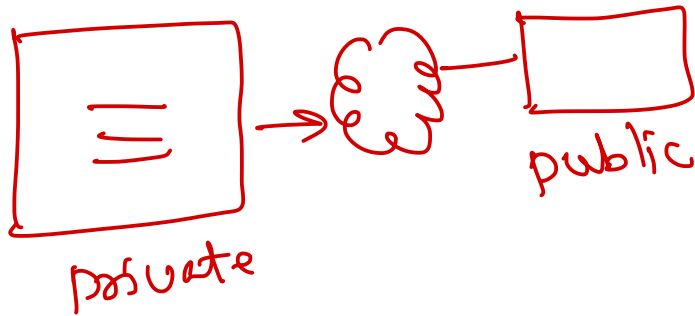
Cloud Deployment Models: Private ✓

- Typically infrastructure used by a single organization
- Such infrastructure may be managed by the organization itself to support various user groups, or it could be managed by a service provider that takes care of it either on-site or off-site
- Private clouds are more expensive than public clouds due to the capital expenditure involved in acquiring and maintaining them
- However, private clouds are better able to address the security and privacy concerns of organizations



Cloud Deployment Models: Hybrid ✓

- Organization makes use of interconnected private and public cloud infrastructure
- Many organizations make use of this model when they need to scale up their IT infrastructure rapidly, such as when leveraging public clouds to supplement the capacity available within a private cloud
- For example, if an online retailer needs more computing resources to run its Web applications during the holiday season it may attain those resources via public clouds.



Cloud Services

→ CPU

- Compute: used to create the Virtual Machine
- Storage: used to provide the storage
- Database: RDBMS + No SQL
- Security and Identity Management
- Media Services
- Machine Learning :
- Cost Management
- Application Integration
 - ↳ SQS, SES, SNS

EC2 - Elastic Compute Cloud → VM → EC2 instance
Storage

- Block Storage : EBS
- File Storage : EFS
- Object Storage : S3 - SSS - Simple Storage Service

Database

↳ RDS

- RDBMS : MySQL, PostgreSQL, Oracle, MS-SQL Server
- NoSQL : DynamoDB, Redis



Advantages

- Lower computer costs (no capex)
- Improved performance
- Reduced software costs
- Instant software updates
- Improved document format compatibility
- Unlimited storage capacity
- Increased data reliability
- Universal document access
- Latest version availability



Disadvantages

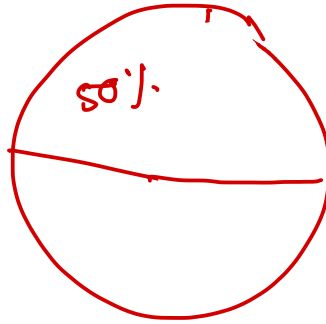
- Requires a constant Internet connection
- Does not work well with low-speed connections
- Features might be limited (depends on the provider)
- Stored data might not be secure (— u —)
- Stored data can be lost (— u —)
- Each cloud systems uses different protocols and different APIs



Cloud Providers

- Amazon Web Services
- Google Cloud Platform
- Microsoft Azure
- Rackspace
- DigitalOcean
- Alibaba Cloud
- Oracle Cloud
- IBM Cloud

⋮

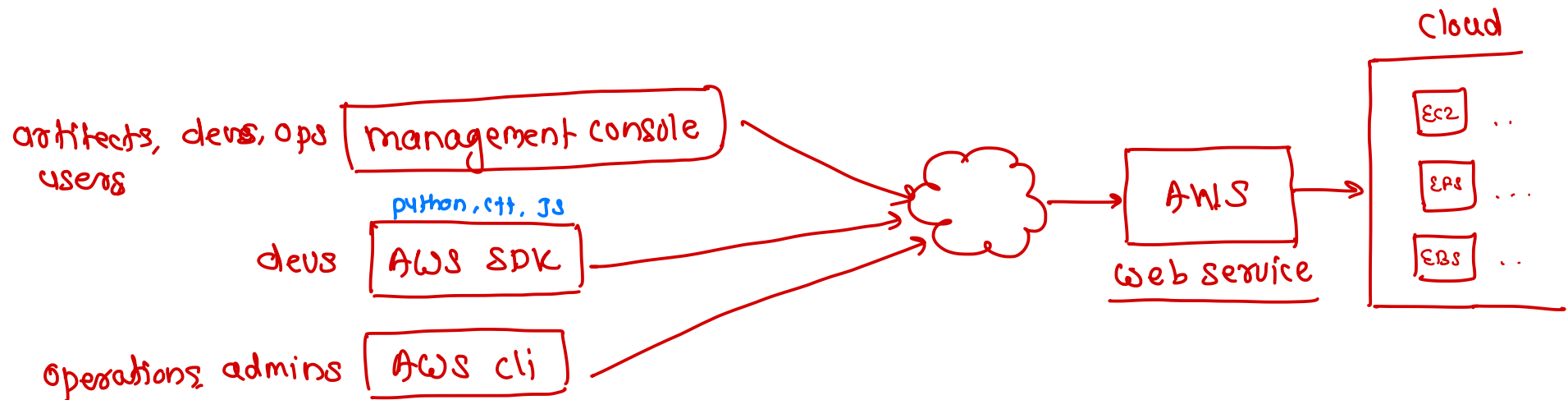


AWS

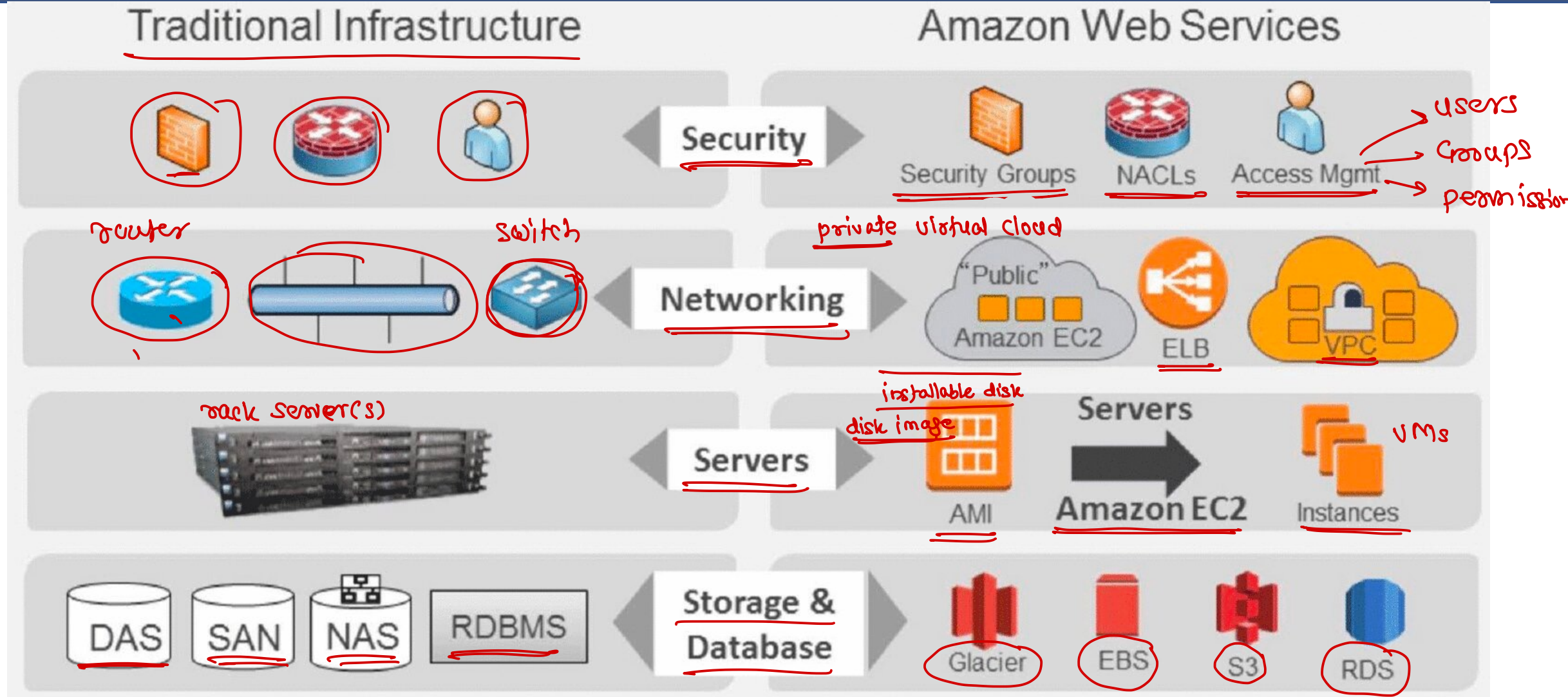


What is AWS ?

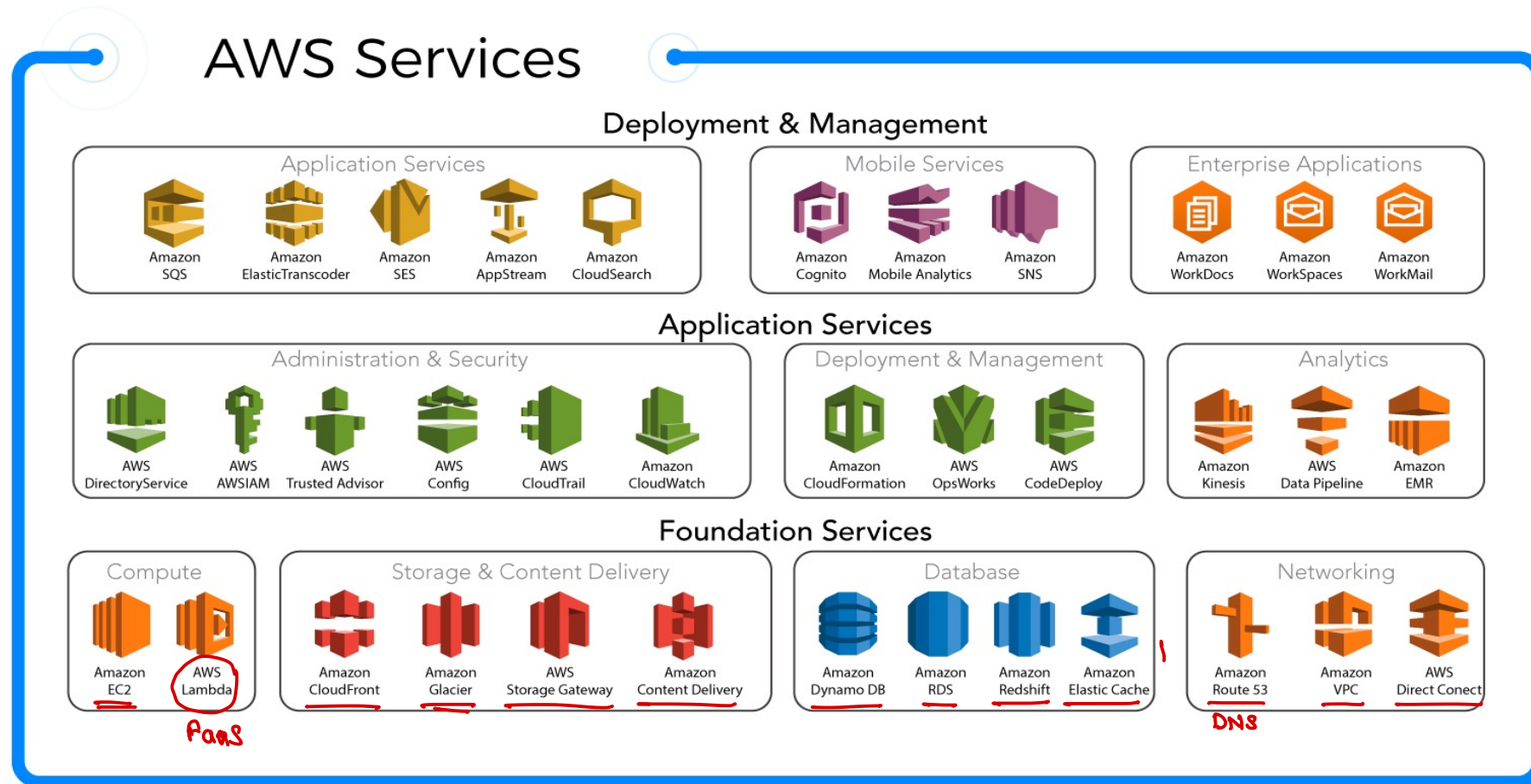
- AWS stands for Amazon Web Services
- Platform that offers flexible, reliable, scalable, easy-to-use and cost-effective cloud computing solutions
- Amazon's cloud implementation
- It's a combination of IaaS, PaaS and SaaS offerings



Traditional vs AWS



AWS Services



Global Infrastructure: Region

- Geographic area having availability zone(s)
- Collection of availability zones that are geographically located close to one other
- Every Region will act independently of the others, and each will contain at least two Availability Zones
- E.g.
 - US East: N. Virginia, Ohio
 - US West: N. California, Oregon
 - Asia Pacific: Mumbai, Seoul, Singapore, Sydney, Tokyo



Global Infrastructure: Availability Zone

- Essentially the physical data centers of AWS
- This is where the actual compute, storage, network, and database resources are hosted that we as consumers provision within our Virtual Private Clouds (VPCs)
- Availability Zones are always referenced by their Code Name, which is defined by the AZs Region Code Name that the AZ belongs to, followed by a letter
- E.g.
 - the AZs within the eu-west-1 region (EU Ireland), are
 - eu-west-1a
 - eu-west-1b
 - eu-west-1c



Global Infrastructure: Edge Locations

- Edge Locations are AWS sites deployed in major cities and highly populated areas across the globe
- Generally used to cache data and reduce latency for end-user access by using the Edge Locations as a global Content Delivery Network (CDN)
- Edge Locations are primarily used by end users who are accessing and using your services
- E.g.
 - Route 53: DNS Lookup
 - CloudFront
 - Content Delivery Network (CDN)
 - Cached contents, streaming distribution, acceleration



EC2



- Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides resizable compute capacity in the cloud
- It is a virtual machine you will be building in the cloud
- EC2 instances are designed to mimic traditional on-premise servers, but with the ability to be commissioned and decommissioned on-demand for easy scalability and elasticity
- EC2 supports variety of operating systems:
 - Linux: Amazon Linux, Ubuntu, Red Hat Enterprise, SUSE Linux Enterprise Server, Fedora, Debian, CentOS, Gentoo Linux, Oracle Linux, FreeBSD
 - Windows: Windows Server, Windows
- Every instance comprised of
 - Amazon Machine Image (AMI)
 - Instance type
 - Network Interface
 - Storage



Step 1 - Choose AMI

Launch instance wizard | EC2

← → ↺ https://console.aws.amazon.com/ec2/v2/home?region=us-east-1#LaunchInstanceWizard: ☆ ASP 📷 🔥 🔒 A ⋮

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1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 1: Choose an Amazon Machine Image (AMI) [Cancel and Exit](#)

 Windows	Microsoft Windows Server 2012 R2 with SQL Server 2016 Enterprise - ami-02c96bdc8f851187b Microsoft Windows Server 2012 R2 Standard edition, 64-bit architecture, Microsoft SQL Server 2016 Enterprise edition. [English] Root device type: ebs Virtualization type: hvm ENA Enabled: Yes	Select 64-bit (x86)
 Windows Free tier eligible	Microsoft Windows Server 2012 Base - ami-0f83e0a40f08208db Microsoft Windows 2012 Standard edition with 64-bit architecture. [English] Root device type: ebs Virtualization type: hvm ENA Enabled: Yes	Select 64-bit (x86)
 Windows Free tier eligible	Microsoft Windows Server 2008 R2 Base - ami-094c0d39ab684c29a Microsoft Windows 2008 R2 SP1 Datacenter edition, 64-bit architecture. [English] Root device type: ebs Virtualization type: hvm ENA Enabled: Yes	Select 64-bit (x86)
 Amazon Linux	Amazon Linux 2 LTS with SQL Server 2017 Standard - ami-0da21cbc0d899ebc7 Microsoft SQL Server 2017 Standard edition on Amazon Linux 2 LTS. The AMI also comes pre-installed with .NET Core 2.0 and PowerShell 6.0. Root device type: ebs Virtualization type: hvm ENA Enabled: Yes	Select 64-bit (x86)
 Ubuntu	Ubuntu Server 16.04 LTS (HVM) with SQL Server 2017 Standard - ami-66ca1419 Microsoft SQL Server 2017 Standard edition on Ubuntu Server 16.04 LTS. The AMI also comes pre-installed with .NET Core 2.0 and PowerShell 6.0. Root device type: ebs Virtualization type: hvm ENA Enabled: Yes	Select 64-bit (x86)
 Amazon Linux Free tier eligible	.NET Core 2.1 with Amazon Linux 2 - Version 1.0 - ami-0e24aee0334f94155 .NET Core 2.1 and the PowerShell 6.0 pre-installed to run your .NET Core applications on Amazon Linux 2 with Long Term Support (LTS). Root device type: ebs Virtualization type: hvm ENA Enabled: Yes	Select 64-bit (x86)
 Ubuntu Free tier eligible	.NET Core 2.1 with Ubuntu Server 18.04 - Version 1.0 - ami-e24b7d9d .NET Core 2.1 and the PowerShell 6.0 pre-installed to run your .NET Core applications on Ubuntu 18.04 with Long Term Support. Root device type: ebs Virtualization type: hvm ENA Enabled: Yes	Select 64-bit (x86)

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Amazon Machine Image (AMI)

- Operating System used to create virtual machine (EC2 instance)
- AMI are built for a specific region
- You can copy an AMI from one region to another
- You can also create a custom AMI with required applications/configuration
- AMI contains
 - Template for root volume
 - Launch permissions that control which account can use the AMI
 - EBS mapping that specifies the volume(s) to attach the instance when its launched
- AMI comes into two types
 - Instance store backed AMI
 - EBS backed AMI



Step 2 - Instance Type

Launch instance wizard | EC2

https://console.aws.amazon.com/ec2/v2/home?region=us-east-1#LaunchInstanceWizard:

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1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

6. Configure Security Group

7. Review

Step 2: Choose an Instance Type

Amazon EC2 provides a wide selection of instance types optimized to fit different use cases. Instances are virtual servers that can run applications. They have varying combinations of CPU, memory, storage, and networking capacity, and give you the flexibility to choose the appropriate mix of resources for your applications. [Learn more](#) about instance types and how they can meet your computing needs.

Filter by:

All instance types

Current generation

Show/Hide Columns

Currently selected: t2.micro (Variable ECUs, 1 vCPUs, 2.5 GHz, Intel Xeon Family, 1 GiB memory, EBS only)

	Family	Type	vCPUs	Memory (GiB)	Instance Storage (GB)	EBS-Optimized Available	Network Performance	IPv6 Support
☐	General purpose	t2.nano	1	0.5	EBS only	-	Low to Moderate	Yes
☒	General purpose	t2.micro Free tier eligible	1	1	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.small	1	2	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.medium	2	4	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.large	2	8	EBS only	-	Low to Moderate	Yes
☐	General purpose	t2.xlarge	4	16	EBS only	-	Moderate	Yes
☐	General purpose	t2.2xlarge	8	32	EBS only	-	Moderate	Yes
☐	General purpose	t3.nano	2	0.5	EBS only	Yes	Up to 5 Gigabit	Yes
☐	General purpose	t3.micro	2	1	EBS only	Yes	Up to 5 Gigabit	Yes
☐	General purpose	t3.small	2	2	EBS only	Yes	Up to 5 Gigabit	Yes
☐	General purpose	t3.medium	2	4	EBS only	Yes	Up to 5 Gigabit	Yes
☐	General purpose	t3.large	2	8	EBS only	Yes	Up to 5 Gigabit	Yes
☐	General purpose	t3.xlarge	4	16	EBS only	Yes	Up to 5 Gigabit	Yes

Cancel

Previous

Review and Launch

Next: Configure Instance Details

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Instance Type

- Used to decide the EC2 instance configuration
- AWS provides various instance types [<https://aws.amazon.com/ec2/instance-types/>]
 - General purpose: A (ARM), T (Cheapest), M (Main)
 - Compute optimized: C (Compute)
 - Memory optimized: R (RAM), X (Extreme RAM), Z (High compute and memory)
 - Accelerated computing: P (Picture-GPU), G (Graphics), F (Fast)
 - Storage optimized: I (IOPS), D (Data), H (High Disk Throughput)



Instance Types

Type	Category	Description	Use Cases
M5	General Purpose	Balance of compute, memory and network resources	Mid-sized databases
C5	Compute Optimized	Advanced CPUs	Modelling, Analytics
H1	Storage Optimized	Local HDD Storage	Map Reduce
R4	Memory Optimized	More RAM for \$	In-memory caching
X1	Memory Optimized	Terabytes of RAM and SSD	In-memory database
I3	IO Optimized	Local SSD storage, high IOPS	NoSQL databases
G3	GPU Graphics	GPUs with video encoders	3d rendering
P3	GPU Compute	GPUs with tensor cores	Machine Learning
F1	Accelerated Computing	FPGA, custom hardware accelerations	Genomics
T2	Burstable	Shared CPUs, lowest cost	Web servers



Step 3 – Configure Instance Details

Launch instance wizard | EC2

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1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 3: Configure Instance Details

Configure the instance to suit your requirements. You can launch multiple instances from the same AMI, request Spot instances to take advantage of the lower pricing, assign an access management role to the instance, and more.

Number of instances ⓘ

1

[Launch into Auto Scaling Group ⓘ](#)

Purchasing option ⓘ

☐ Request Spot instances

Network ⓘ

vpc-e658e69c | Default (default) ⌵ ⌵

[Create new VPC](#)

Subnet ⓘ

No preference (default subnet in any Availability Zone) ⌵ ⌵

[Create new subnet](#)

Auto-assign Public IP ⓘ

Use subnet setting (Enable) ⌵ ⌵

Placement group ⓘ

☐ Add instance to placement group

Capacity Reservation ⓘ

Open ⌵ ⌵

[Create new Capacity Reservation](#)

IAM role ⓘ

None ⌵ ⌵

[Create new IAM role](#)

Shutdown behavior ⓘ

Stop ⌵ ⌵

Enable termination protection ⓘ

☐ Protect against accidental termination

Monitoring ⓘ

☐ Enable CloudWatch detailed monitoring

[Additional charges apply.](#)

Tenancy ⓘ

Shared - Run a shared hardware instance ⌵ ⌵

[Additional charges will apply for dedicated tenancy.](#)

Elastic Inference ⓘ

☐ Add an Elastic Inference accelerator

[Additional charges apply.](#)

T2/T3 Unlimited ⓘ

☐ Enable

[Additional charges may apply](#)

▶ Advanced Details

Cancel

Previous

Review and Launch

Next: Add Storage

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On Demand

- Low cost and flexibility with no upfront cost
- Pay for hours used with no commitment
- Good option for developers or testers
- Ideal for
 - auto scaling groups
 - Unpredictable workloads



Step 4 – Add Storage

Launch instance wizard | EC2

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Resource Groups

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1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

6. Configure Security Group

7. Review

Step 4: Add Storage

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes. [Learn more](#) about storage options in Amazon EC2.

Volume Type ⓘ	Device ⓘ	Snapshot ⓘ	Size (GiB) ⓘ	Volume Type ⓘ	IOPS ⓘ	Throughput (MB/s) ⓘ	Delete on Termination ⓘ	Encrypted ⓘ
Root	/dev/sda1	snap-0ac9e6b9d926e36c9	8	General Purpose SSD (gp2)	100 / 3000	N/A	☒	Not Encrypted

Add New Volume

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage. [Learn more](#) about free usage tier eligibility and usage restrictions.

Cancel

Previous

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Next: Add Tags

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Storage options

- Block storage
 - Instance store
 - EBS volume
- Object storage
 - S3
- File sharing
 - EFS



EBS – Solid State Drives (SSD)

- General Purpose (gp2)
 - Used for dev/test environments
 - Baseline performance: 100 IOPS – 3000 IOPS
 - Bursts up to 32000 IOPS (credit based)
 - Volume size from 1GiB to 16 TiB
- Provisioned IOPS (io1)
 - Used for mission critical applications that require sustained IOPS performance
 - Large database workloads
 - Volume size: 4GiB to 16TiB
 - Performs at provisioned level and can provision up to 64K IOPS per volume
 - NOTE: EC2 instances have a maximum of 80K IOPS total



EBS – Hard Disk Drive (HDD)

- Can not be boot volume
- Size: 500GiB – 16TiB
- Types
 - Throughput Optimized (st1)
 - Low storage cost
 - Used for frequently accessed, throughout intensive workloads
 - E.g. Streaming, Big Data
 - Max throughput of 500 MiB/s
 - NOTE: EC2 instances have a maximum throughput of 1750MiB/s
 - Cold HDD (sc1)
 - Lowest cost
 - Infrequently accessed data
 - Max throughput of 250MiB/s



Step 5 – Add Tags

Launch instance wizard | EC2

+

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Resource Groups ▾

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Support ▾

1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

6. Configure Security Group

7. Review

Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.

A copy of a tag can be applied to volumes, instances or both.

Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

Key (127 characters maximum)	Value (255 characters maximum)	Instances ⓘ	Volumes ⓘ
This resource currently has no tags			
Choose the Add tag button or click to add a Name tag .			
Make sure your IAM policy includes permissions to create tags.			

Add Tag

(Up to 50 tags maximum)

Cancel

Previous

Review and Launch

Next: Configure Security Group

Feedback

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Step 6 - Add Security Group

Launch instance wizard | EC2

https://console.aws.amazon.com/ec2/v2/home?region=us-east-1#LaunchInstanceWizard:

☆

ASP

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Support

1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

6. Configure Security Group

7. Review

Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group: ☒ Create a new security group
☐ Select an existing security group

Security group name:

Description:

Type	Protocol	Port Range	Source	Description
SSH	TCP	22	Custom 0.0.0.0/0	e.g. SSH for Admin Desktop

Add Rule

Warning

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel

Previous

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Security Group

- Acts as a virtual firewall for your instance to control inbound and outbound traffic
- Controls the ports and protocols that can reach the front-end listener
- Every EC2 instance must have at least one security group attached
- Up to 5 security groups can be attached to an EC2 instance
- Security groups act at the instance level, not the subnet level
- Security group contains rules
 - You can specify allow rules, but not deny rules
 - You can specify separate rules for inbound and outbound traffic
 - When you create a security group, it has no inbound rules
 - By default, a security group includes an outbound rule that allows all outbound traffic
 - Security groups are stateful
 - Instances associated with a security group can't talk to each other unless you add rules allowing it
 - Security groups are associated with network interfaces



Step 7 – Review and create PEM file

The screenshot shows the AWS Management Console at the 'Launch instance wizard' page, specifically Step 7: Review Instance Launch. The page has a breadcrumb trail: 1. Choose AMI, 2. Choose Instance Type, 3. Configure Instance, 4. Add Storage, 5. Add Tags, 6. Configure Security Group, 7. Review. A warning message at the top states: 'Improve your instances' security. Your security group, launch-wizard-6, is open to the world. Your instances may be accessible from any IP address. We recommend that you update your security group rules to allow access from known IP addresses only. You can also open additional ports in your security group to facilitate access to the application or service you're running, e.g., HTTP (80) for web servers. Edit security groups'. The main content area is divided into sections: AMI Details (Ubuntu Server 16.04 LTS (HVM), SSD Volume Type - Free tier eligible), Instance Type (t2.micro, Variable ECUs, 1 vCPU, 1 Memory), Security Groups (launch-wizard-6), and Instance Details. A modal dialog titled 'Select an existing key pair or create a new key pair' is open in the center. The dialog contains the following text: 'A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance. Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about removing existing key pairs from a public AMI.' Below the text are two dropdown menus: 'Choose an existing key pair' and 'Select a key pair' (with 'cloud-workshop' selected). There is also a checkbox: 'I acknowledge that I have access to the selected private key file (cloud-workshop.pem), and that without this file, I won't be able to log into my instance.' At the bottom of the dialog are 'Cancel' and 'Launch Instances' buttons. The background page shows a table for Security Groups with columns for Type, Protocol, Port Range, and CIDR IP ranges. The bottom of the page has a footer with 'Feedback', 'English (US)', and copyright information.

Launch instance wizard | EC2 | x +

https://console.aws.amazon.com/ec2/v2/home?region=us-east-1#LaunchInstanceWizard:

Services Resource Groups

1. Choose AMI 2. Choose Instance Type 3. Configure Instance 4. Add Storage 5. Add Tags 6. Configure Security Group 7. Review

Step 7: Review Instance Launch

Please review your instance launch details. You can go back to edit changes for each section. Click **Launch** to assign a key pair to your instance and complete the launch process.

⚠ Improve your instances' security. Your security group, launch-wizard-6, is open to the world.
Your instances may be accessible from any IP address. We recommend that you update your security group rules to allow access from known IP addresses only. You can also open additional ports in your security group to facilitate access to the application or service you're running, e.g., HTTP (80) for web servers. [Edit security groups](#)

AMI Details

Ubuntu Server 16.04 LTS (HVM), SSD Volume Type - Free tier eligible

Instance Type

Instance Type	ECUs	vCPUs	Memory (GB)
t2.micro	Variable	1	1

Security Groups

Security group name	Description
launch-wizard-6	launch-wizard-6 created 2019-04-01T...

Instance Details

Storage

Tags

Select an existing key pair or create a new key pair

A key pair consists of a **public key** that AWS stores, and a **private key file** that you store. Together, they allow you to connect to your instance securely. For Windows AMIs, the private key file is required to obtain the password used to log into your instance. For Linux AMIs, the private key file allows you to securely SSH into your instance.

Note: The selected key pair will be added to the set of keys authorized for this instance. Learn more about [removing existing key pairs from a public AMI](#).

Choose an existing key pair

Select a key pair

cloud-workshop

☐ I acknowledge that I have access to the selected private key file (cloud-workshop.pem), and that without this file, I won't be able to log into my instance.

Cancel Launch Instances

Feedback English (US)

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EC2 Key Pairs

- Uses PEM format (Privacy Enhanced Mail)
- Used to authenticate a client when logging into EC2 instance
- Each key pair consists of a public key and a private key
- AWS stores the public key on the instance and you are responsible for storing the private key
- To log into the instance you must create and authenticate with key pair
 - Linux instances have no password and you use a key pair to log in
 - With windows you use a key pair to obtain the administrator password and then log into the instance with RDP
- During the creation process of an EC2 instance you are required to either create a new key pair or use existing pair
- The private key is available for download and stored on your local drive
- NOTE: it will be available only once in the form of .pem file



Linux Programming and Cloud Computing

- Objective: To introduce Linux environment and hands on Linux commands.

Contents

- Linux Architecture and Structure
- Basic commands
- Advanced commands
- vi/vim editor
- Linux shell programming (shell scripts)

Evaluations

- Theory - 40 marks
- Lab - 40 marks
- Internal - 20 marks

What is OS?

- Interface between end user and computer hardware.
- Interface between Programs and computer hardware.
- Control program that controls execution of all other programs. * Resource manager/allocator that manage all hardware resources.
- Bootable CD/DVD = Core OS + Applications + Utilities
- Core OS = Kernel -- Performs all basic functions of OS.

OS Functions

Process Management

- Process creation/termination.
- Process life cycle.
- Inter-process communication.
- Multi-threading. Synchronization.
- Deadlock.

CPU scheduling

- Scheduling criteria
- Scheduling types
- Scheduling algorithms.

Memory Management

- Contiguous allocation
- Segmentation Paging
- Virtual memory

- Demand Paging
- Page replacement algorithms

File & IO Management

- File and File-Systems
- Disk allocation algorithms
- Disk scheduling algorithms
- Interrupts

Hardware abstraction

- Hides hardware intricacies from end user and applications

User interfacing

- Shell (GUI/CLI)

Security & Protection

- Securing system/applications from internal and external threats

Networking

- Communication between two computers

Process Management

Program

- Set of instructions given to the computer --> Executable file.
- Program --> Sectioned binary --> "objdump" & "readelf".
 - Exe header --> Magic number, Address of entry-point function, Information about all sections. (objdump -h program.out)
 - Text --> Machine level code (objdump -S program.out)
 - Data --> Global and Static variables (Initialized)
 - BSS --> Global and Static variables (Uninitialized)
 - RoData --> String constants
 - Symbol Table --> Information about the symbols (Name, Size, section, Flags, Address) (objdump -t program.out)
- Program (Executable File) Format
 - Windows -- PE
 - Linux -- ELF
- Program are stored on disk (storage).

Process

- Program under execution
- Process execute in RAM.
- Process control block contains information about the process (required for the execution of process).

- Process id
- Exit status
 - 0 - Indicate successful execution
 - Non-zero - Indicate failure
- Scheduling information (State, Priority, Sched algorithm, Time, ...)
- Memory information (Base & Limit, Segment table, or Page table)
- File information (Open files, Current directory, ...)
- IPC information (Signals, ...)
- Execution context (Values of CPU registers)
- Kernel stack
- PCB is also called as process descriptor (PD), uarea (UNIX), or task_struct (Linux).
- In Linux size of task_struct is approx 4KB

User interfacing

- UI of OS is a program (Shell) that interface between End user and Kernel.
- Shell -- Command interpreter
 - End user --> Command --> Shell --> Kernel
- User interfacing (Shell)
 - Graphical User Interface (GUI)
 - Command Line Interface (CLI)

Example shells

- Windows
 - GUI shell: explorer.exe
 - CLI shell: cmd.exe, powershell.exe
- DOS
 - CLI shell: command.com
- Unix/Linux
 - CLI shell: bsh, "bash", ksh, csh, zsh, ...
 - ls /bin/*sh
 - echo \$SHELL
 - shell of current user can be changed using "chsh" command.
- GUI shell/standards
 - GNOME: GNU Network Object Model Environment (e.g. Ubuntu, Redhat, CentOS, ...)
 - KDE: Kommon Desktop Environment (e.g. Kubuntu, SuSE, ...)

Disk/partition naming conventions

- * Windows:
 - * Disks are named as disk0, disk1, ...
 - * partitions are named as drives i.e. C:, D:, E:, ...
- * Linux:
 - * Disks are named as /dev/sda, /dev/sdb, /dev/sdc, etc.
 - * Partitions per disk are named as

```
* sda partitions: sda1, sda2, sda3, ...  
* sdb partitions: sdb1, ...
```

Linux File Structure

- Linux follows "/" (root) file system.
- "/" is a starting point of Linux file system.
- All your data is stored in this partition.
- / contains boot, bin, sbin, etc, root, home, dev, proc, mnt, media, opt
- In Linux everything is a file.
- Mainly there are two types of files in Linux
 - File
 - Directory (Folder)
- Linux Directories
 - boot - files related to booting
 - vmlinuz - kernel Image
 - grub - boot loader
 - config - kernel configuration
 - initrd/initramfs - initail root file system
 - bin - user commands in binary format
 - sbin - all admin/system commands in binary format
 - etc - configuration files
 - root - home directory of root user
 - home - it contains sub directories for each user with its name
 - devendra -> /home/devendra
 - sunbeam -> /home/sunbeam
 - osboxes -> /home/osboxes
 - dev - it contains all device related files
 - lib - shared program libraries required by kernel
 - mnt - it is temporary mount point
 - media - it is mount point for media eg cdrom
 - opt - stores optional files of large softwares
 - proc - virtual file system - it contains information about system or processes
 - sys - entries of each block devices, subdirectories for each physical bus type supported, every device class registered with the kernel, global device hierarchy of all devices
 - tmp - temporary files that may be lost on system shutdown
 - usr - read only directory that stores small programs and files accessible to all users

Path

- It is a unique location of any file in the file system.
- It is represented by character strings with few delimiters ("/", "\", ".")
- Types of path
 - There are two types of paths in linux
 - Absolute path

- Path which starts with "/" is called as absolute path.
- E.g. /home/devendra/MyData/Demos/demo01.sh
- Relative path
 - Path with respect to current directory is called as relative path
 - E.g. MyData/Assignments/assign02.pdf

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File Management

File

- File is collection of data/information on storage device.
 - File = Contents (Data) + Information (Metadata)
 - The data is stored in zero or more Data blocks (in FS), while metadata is stored in the FCB (in filesystem).
- FCB is called as "inode" on UNIX/Linux. It contains
 - type: UNIX/Linux has 7 types of files
 - -: regular, d: directory, l: symbolic link, p: pipe, s: socket, c: char device, b: block device
 - size: number of bytes
 - links: number of hard links
 - mode (permissions): (u) rwx, (g) rwx, (o) rwx
 - user & group
 - time-stamps: modification, creation, access.
 - info about data blocks
- terminal> ls -l
 - type, mode, links, user, group, size, timestamp, name.
- terminal> stat filepath

File System

- Files are stored on storage device. Arrangement of files in storage device is called as "File System".
- e.g. FAT, NTFS, EXT2/3/4, ReiserFS, XFS, HFS, etc.
- File System logically divide partition into 4 sections.
 - Boot block/Boot sector
 - Contains programs/info required for booting of OS
 - Typically contains bootstrap program and bootloader program
 - Super block/Volume control block
 - Contains information of whole partition.
 - Capacity, Label.
 - terminal> df -h
 - Total number of data blocks/inodes.
 - Number of used/free data blocks/inodes.
 - Information of free data blocks/inodes.
 - Inode List/Master file table
 - Inodes (FCB) for each file
 - Data blocks
 - Stores data of the file.
 - Each file have zero or more data blocks.
 - Size of data blocks can be configured while creating file system
- File system is created by the format utility while formatting the partition.

- Windows: format.exe
- Linux: mkfs
 - terminal> sudo mkfs -t ext3 /dev/sdb1
 - terminal> sudo mkfs -t vfat /dev/sdb1
 - -t fs_type e.g. ext3, ext4, vfat, ntfs, ...
 - partition e.g. /dev/sdb1

Links

- Links are shortcuts to access deeply located files quickly.
- There are two types of links in Linux/UNIX.
 - 1. Symbolic Link
 - 2. Hard Link

Symbolic Link

- terminal> ln -s /path/of/target/file linkpath
- Internally use symlink() syscall.
 - int symlink(const char *target_path, const char *link_path);

symlink() syscall

- A new link file is created (new inode and new data block is allocated), which contains info about the
 - target file (absolute or relative path).
- Link count is not incremented.
- If target file is deleted, the link becomes useless.
- Can create symlinks for directories also.

Hard Link

- terminal> ln targetfilepath linkfilepath
- Internally use link() syscall.
 - int link(const char *target_path, const char *link_path);

link() syscall

- A new directory entry is created, which has a new name and same inode number. No new file (inode and data blocks) is created.
- Link count in the inode of the file is incremented.
- If directory entry of target file is deleted (rm command), file can be still accessed by link directory entry.
- Cannot create hard link for directories, because it may lead to infinite recursion (while traversing directories recursively e.g. ls -R)

Regular Expressions

- Find a pattern in text file(s).
- Regular expressions are patterns used to match character combinations in strings.

- A regular expression pattern is composed of simple characters, or a combination of simple and special characters e.g. /abc/, /ab*c/
- Pattern is given using regex wild-card characters.
 - Basic wild-card characters
 - \$ - find at the end of line.
 - ^ - find at the start of line.
 - [] - any single char in give range or set of chars
 - [^] - any single char not in give range or set of chars
 - . - any single character
 - * - zero or more occurrences of previous character
 - Extended wild-card characters
 - ? - zero or one occurrence of previous character
 - + - one or more occurrences of previous character
 - {n} - n occurrences of previous character
 - {n,} - max n occurrences of previous character
 - {m,} - min m occurrences of previous character
 - {m,n} - min m and max n occurrences of previous character
 - () - grouping (chars)
 - (|) - find one of the group of characters

grep

- Regex commands
 - grep - GNU Regular Expression Parser - Basic wild-card
 - egrep - Extended Grep - Basic + Extended wild-card
 - fgrep - Fixed Grep - No wild-card
- Command syntax
 - grep "pattern" filepath
 - grep [options] "pattern" filepath
 - -c : count number of occurrences
 - -v : invert the find output
 - -i : case insensitive search
 - -w : search whole words only
 - -R : search recursively in a directory
 - -n : show line number.
- Anchoring - ^, \$
- Single Character Matching - .
- Bracket Expressions - [], [^]
- Quantifiers - *, ?, +, {n}, {n,}, {n}, {m,n}
- Alternation - |

- Grouping - ()

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Command Nesting

- `cmd1 | cmd2`
 - two commands are executed separately
 - output of first command(`cmd1`) is given to the second command(`cmd2`)
- `cmd1 && cmd2`
 - two commands are executed separately
 - output of first command(`cmd1`) is not given to the second command(`cmd2`)
 - if first command (`cmd1`) is failed, then second command (`cmd2`) is not executed
- `cmd1 || cmd2`
 - two commands are executed separately
 - output of first command(`cmd1`) is not given to the second command(`cmd2`)
 - if first command (`cmd1`) is successful, then second command (`cmd2`) is not executed
- `cmd1 ; cmd2`
 - two commands are executed separately
 - output of first command(`cmd1`) is not given to the second command(`cmd2`)

Synchronous and Asynchronous command execution

- Synchronous
 - shell is waiting to complete the command
- Asynchronous
 - shell is not waiting to complete the command

Types of Linux commands

Internal commands

- * these commands(code) are part of shell code
- * manual pages are not available for these commands
- * `help` - list of all builtin commands
- * `help <cmd>` - show the help for given command

External commands

- * separate executable is available for these commands
- * manual pages are only available for these commands

chmod

```
* chmod - used to change the mode bit
* + : add the mode bit
* - : remove the mode bit
* u : user/owner
* g : group
* o : others

* +x : add execute permission to all
* -x : remove execute permission of all
* +r : add read permission to all
* -r : remove read permission of all
* +w : add write permission to user and group
* -w : remove write permission of user and group
* u/g/o + r/w/x : add permissions to specific one
* u/g/o - r/w/x : remove permissions to specific one
```

VI Editor

```
* command line editor
* VIM - Vi IMproved
* To install vim editor
  * $ sudo apt-get install vim
* This editor works in three modes
  * command mode - you can execute commands of vim editor (default mode)
    * press Esc
  * insert mode - you can write/edit the contents of file
    * press i
  * visual mode - you can select the text
    * press v
* To create/edit file into vim editor
  * $ vim <filename>
```

VIM Editor commands

- :w - to save/write into file
- :q - to quit/exit from the editor
- :wq - to save and quit
- u - to undo
- :y - to copy current line
- :ny - to copy nth line of file
- yy - to copy current line

- nyy - to copy n lines from current line
- :m,ny - to copy lines from m to n
- p - to paste the copied line (cursor should be on one line before)
- :d - to cut current line
- :nd - to cut nth line of file
- dd - to cut current line
- ndd - to cut n lines from current line
- :m,nd - to cut lines from m to n
- p - to paste the copied line (cursor should be on one line before)
- :d - to delete current line
- :nd - to delete nth line of file
- dd - to delete current line
- ndd - to delete n lines from current line
- :m,nd - to delete lines from m to n
- yw - copy from current position upto next word
- yiw - copy current word
- y\$ - copy from cursor position upto end of line
- y^ - copy from cursor position upto start of line
- :next - to go into next file
- :prev - to go into previous file
- :wqa - to save, quit all
- vim -o - to open multiple files horizontally
- vim -O - to open multiple files vertically
- ctrl + ww - to go into next file
- /pattern - to search the pattern
- n - to go on next occurrence
- 😞/pattern1/pattern2/ - find and replace only first occurrence of current line
- 😞/pattern1/pattern2/g - find and replace all occurrences of current line

- `:%s/pattern1/pattern2/g` - find and replace all occurrences of file

Shell Scripting/Programming

- Shell Script - set of commands with some programming constructs, written for shell
- commands - shell commands
- Programming construct - control flow structures, functions, arrays
 - Decision control structure (if else)
 - Selection control structure (case)
 - Loop control structure (while, until, for)
- Shell scripts are interpreted by shell
- Shell scripts are used by administration people for installations, setups, automations
- Procedure to write and execute scripts
 - Step 1 - create file with extension .sh
 - `$ vim script.sh`
 - Step 2 - execute the script
 - Option 1
 - `$ bash script.sh`
 - Option 2
 - `$ chmod +x script.sh`
 - `$ ./script.sh`
- echo
 - `-n` - to suppress the new line character at the end of msg
 - `-e` - to enable escape sequences

Operators

- Arithmetic - +, -, /, *, %
- Relational - -eq, -ne, -lt, -gt, -le, -ge
- Logical - -a, -o, !

condition

- test condition
- [condition]

if else

```
if [ condition ]
then
    ...
fi

if [ condition ]
then
    ...
else
    ...
fi

if [ condition ]
then
    ...
elif [ condition ]
then
    ...
else
    ...
fi
```

case

```
case $var in
1)
    ...
;;
2)
    ...
;;
*)
    ...
;;
esac
```

Loops

while

```
initialization
while [ condition ]
do
    body
    modification
done
```

until

```
initialization
until [ condition ]
do
    body
    modification
done
```

for

```
for(( init ; cond ; mod ))
do
    body
done

for variable in [collection]
do
    body
done
```

select

```
select variable in [collection]
do
    body
done
```

File conditionals

- `-e $path` -> true if path exist

- -f \$file -> true if file is regular file
- -d \$file -> true if file is directory
- -w \$file -> true if file has write access
- -r \$file -> true if file has read access
- -x \$file -> true if file has execute access

Positional Parameters

- these are values which are passed to the script from command line

Bootting

- Process from computer power on to OS startup.

Bootable device

- Storage device (disk, pen drive, CD/DVD, etc.) whose boot block contains bootstrap program, is said to be Bootable device.

Bootstrap program

- Bootstrap program can load OS kernel into RAM and start its execution.
- Bootstrap program is different for each OS each version.
- Bootstrap is located in first sector (512 bytes) of a disk/partition.

Bootloader program

- Bootloader program can be configured to show multiple boot options to end user.
- Depending on user selection, it runs Bootstrap program of corresponding OS.
- There are many important bootloader programs.
 - ntldr (Before Windows Vista) --> boot.ini
 - "bootmgr" (Windows Vista Onwards) --> bcd (boot config data)
 - (admin command prompt) cmd> bcdedit
 - LiLo --> Linux Loader
 - "GrUB" (Grand Unified Boot Loader) --> menu.lst or grub.cfg
 - BTX (BooT eXtended) --> BSD Unix
 - SLO (Sparc Interactive Loader) --> Solaris
 - Bootcamp --> Mac OS X
 - "uBoot" --> Linux bootloader for Embedded
- The Bootloader has some config file associated with it.

Bootstrap loader

- The set of programs fixed in Base ROM (Motherboard) is called as "Firmware".
 - BIOS (Basic Input Output System) -- Firmware developed for PC by IBM + MS.
 - EFI (Extensible Firmware Interface) -- Firmware developed for PC by Intel + HP.
- Bootstrap loader is a program from Base ROM.
- It find the bootable device as per "boot device priority" in the BIOS setup.
- Once bootable device is found, it starts its bootloader.

Booting steps

- Computer power on.
- Load firmware (BIOS or EFI) programs from base ROM into RAM.
- Run POST/BIST (from firmware).
- Run Bootstrap loader (from firmware) to find bootable device.
- Bootstrap loader loads bootloader program from bootable device.
- Bootloader program shows multiple options to end user and user select one of them.
- Bootloader program run corresponding bootstrap program.
- Bootstrap program load OS kernel into main memory and OS boot

Linux Booting

- Firmware is loaded in RAM.
- POST is done.
- Bootstrap Loader program finds bootable device.
- Start Bootloader (GrUB):
 - Can start any Linux/OpenSource kernels & chainload to proprietry OS bootloaders.
 - Grub Stage1:
 - Boot Block -- 1st 440 Bytes : Bootstrap (Stage1)
 - It loads stage 1.5.
 - Grub Stage1.5:
 - In first 32 KB of disk.
 - Size approx: 25 KB.
 - It contains basic FS drivers.
 - It loads stage 2.
 - Grub Stage2:
 - In linux boot (/boot) partition
 - Reads config file (grub.cfg)
 - Presents UI for selecting OS to boot (if needed)
 - Loads Linux kernel (vmlinuz) -- self extracting.
- Startup Process
 - Linux kernel access Root file system.
 - systemd/init (sbin directory) process is executed (pid=1).
 - systemd is designed to start (independent) services parallely. In init services were started sequentially.
 - It starts multiple child processes as per configed run-level.
 - Runlevels are like "checkpoints"/"states".
- Runlevels
 - 1 - Single user mode (Used for failsafe/rescue)
 - 2 - Multi user mode (User login enabled, User files accessible)
 - 3 - Networking (Multiuser with Networking on CLI) -- multi-user.target

- 4 - Reserved (unused)
- 5 - Graphical user interface -- graphical.target
- <https://likegeeks.com/linux-runlevels/>
- Linux commands
 - terminal> runlevel
 - terminal> init 0 --> shutdown
 - terminal> init 6 --> reboot
 - terminal> init 3 --> go to runlevel 3 (GUI will be stopped if already started)
 - terminal> init 5 --> go to runlevel 5 (GUI will be started)
 - terminal> startx --> start GUI (give this command from runlevel 3)
- During booting runlevel can be given as bootparam (in GrUB -- edit entry before booting).
 - Press "e" to edit the entry.
 - linux /boot/vmlinuz-x.y.z root=UUID=xxxxxxxxxxxxxx ro splash=no init 3

diff

- diff file1.txt file2.txt
 - use symbols : a-add, c-change, d-delete
 - 0a1 - line1 from file2 is to be added after line0 in file1
 - 2,4c2 - line 2 to 4 of file1 need to change to match line 2 of file2
 - 2,4d2 - line 2 to 4 of file1 need to delete so that both files sync up at line 2
- diff -c file1.txt file2.txt
 - '<' - lines from first file
 - '>' - lines from second file
 - "---" - separates lines of file1 and file2
 - '-' - indicate lines in the first file to be deleted
 - '+' - indicates lines in second file to be added to the first file
- diff -y file1.txt file2.txt

chown

- sudo chown user file
- sudo chown :group file
- sudo chown user:group file
- sudo chown -R user file
- sudo chown -h user file

chgrp

- chgrp group file

Group's commands

- groups
- groups username

- `sudo addgroup groupname`
- `sudo delgroup groupname`
- `sudo usermod -a -G groupname username`
- `sudo usermod -g groupname username` (changes the primary group of user)

User's commands

- `sudo adduser username`
- `sudo adduser -g pgroupname -G wheel,developers username`
- `sudo passwd username`
- `sudo deluser username`
- `sudo deluser --remove-home username`
- `sudo deluser -r username`

mount and unmount

- Mounting enable us to access file system contents on a device/partition under another (current) file system.
- The mount command mounts given device partition to given mount point directory.
 - device partition e.g. `/dev/sdb1`
 - mount point is an empty directory under current file system. e.g. `/mnt`. * After mounting, device contents will be visible under that directory. fs type can also be specified while mounting. If not given, it will be auto-detection.
- `sudo mount -t vfat /dev/sdb1 /mnt`
- `sudo umount /mnt`

tar

- `tar cvf file.tar *.txt`
- `tar xvf file.tar`
- `tar cvzf file.tar.gz *.txt`
- `tar cvzf file.tar.gz`

zip and unzip

- `zip file.zip filename.txt`
 - `-d` - removes file from zip
 - `-u` - updates file to zip
 - `-r` - zip directory recursively
- `unzip file.zip`

gzip and gunzip

- `gzip file.txt`

- -f - to force gzip to do
 - -k - to keep original file
- gzip -r directory
- -1 to -9 level of compression
- gzip -d file.txt.gz (decompress)
- gunzip file.txt.gz

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